Do Felon Voting Rules Affect Criminal Prosecutions?

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Abstract

Felon disenfranchisement laws can alter the composition of the electorate. Accordingly, we argue that in states with such laws, elected prosecutors may be tempted to assign felon status strategically. We test our argument using criminal case data from Virginia and leveraging a sudden policy change — unexpectedly announced by the Governor on April 22, 2016 — enfranchising all felons upon completion of their prison or probation sentence. Using well-known racial differences in partisanship as a measure of likely political alignment, we show that the announcement corresponded to an *increase* in criminal charge severity and sentence length for defendants who were politically unaligned with their prosecutors, and a *decrease* for aligned defendants. Evidence suggests that these changes may have been partially motivated by partisan concerns in the lead-up to the 2016 federal elections.

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In recent years, scholars have gained new insight into the American justice system by studying criminal justice officials, such as judges and prosecutors, as political agents motivated by the desire to remain in office. Focusing especially on the incentives of elected judges and prosecutors, scholars have shown that these officials may manipulate criminal charges and sentences in order to signal ideological congruence or competence to voters (e.g., Gordon and Huber, 2002, 2007; Stuntz, 2004; Dyke, 2007; Bandyopadhyay and McCannon, 2014; Gordon and Yntiso, 2022). These results are consistent with a vast formal and empirical literature on political accountability, in which political agents—whether bureaucrats accountable to elected officials (e.g., Gailmard and Patty, 2007, 2012), or elected officials accountable to the public (e.g., Canes-Wrone, Herron and Shotts, 2001; Alt, Bueno de Mesquita and Rose, 2011)—must make policy choices with an eye toward the principals who keep them in office.[1]

In this paper, we argue that elected criminal justice officials are similar to other political actors not merely in having a principal-agent relationship with an electorate, but also in having powers, vis-a-vis their principals, which ordinary agents lack. For example, rather than choosing a set of policies her constituents approve, a legislator may try to choose a new set of constituents which approves her policies, via gerrymandering or the manipulation of voting restrictions such as felon disenfranchisement laws (Bateman, 2018; Helmke, Kroeger and Paine, 2022; Eubank and Fresh, 2022).[2] We argue that elected criminal justice officials may similarly be able to shape the composition of their local electorates to favor themselves, their allies, and their preferred political parties, and disfavor their opponents. The reason is that many states in the U.S. restrict felon voting, allowing officials to shape the electorate through the selective imposition of disenfranchising felony sentences. The temptation to do this may be especially great for elected

[1]For more comprehensive reviews of this literature, see Ashworth (2012); Gailmard (2014).

[2]Or, in some cases, they may render the composition of the constituency irrelevant by simply replacing elected positions with appointed ones (Komisarchik, 2026).

prosecutors, whose immense discretion gives them an outsize role in the administration of justice (Gordon and Huber, 2009; Stuntz, 2011; Tonry, 2012).

To illustrate this argument, we construct a simple formal model featuring a prosecutor who may offer a defendant a plea deal or go to trial, and a defendant who may accept or reject the plea. Trial is costly to both players, and the prosecutor wants to secure a conviction and as high a sentence as possible, while the defendant wants to minimize her expected loss. We show that when the prosecutor also stands to benefit from the outcome of an upcoming election, and there is a felon disenfranchisement law on the books, the prosecutor's optimal plea changes depending on (1) whether the prosecutor wants the defendant to vote, and (2) whether disenfranchisement is indefinite or only for the duration of the felony sentence. If the prosecutor does *not* want the defendant to vote, we show that when disenfranchisement is indefinite, his offer is more lenient (to increase the probability that the defendant accepts), and when it is sentence-dependent, his offer is harsher (to increase the length of time the defendant is disenfranchised). If the prosecutor *does* want the defendant to vote, the converse is true. In a 'hybrid' regime where with some probability disenfranchisement is indefinite and with reciprocal probability it is sentence-dependent, any increase in the probability that it is sentence-dependent increases the harshness of the optimal plea when the prosecutor does not want the defendant to vote, and decreases it when he does.

To investigate the empirical validity of our argument, we leverage an unanticipated shock to the relationship between felony criminal sentencing and felon disenfranchisement in the state of Virginia: Governor Terry McAuliffe's unexpected 2016 announcement of the automatic enfranchisement of all felons upon completion of their "sentences of [felony] incarceration" and "supervised release." This announcement heralded a substantial change from the previous regime, in which the length of disenfranchisement and the procedures for rights restoration varied across different types of felony convictions (approximating the 'hybrid' regime in our model). It had several immediate consequences. First, it made the length of a felony prison or probation sentence the key determinant of (dis)enfranchisement going forward.[3] Second, according to the governor, it resulted in the immediate enfranchisement of approximately 200,000 individuals (Horwitz and Portnoy, 2016). Third, it motivated voter registration drives across the state, targeting the newly enfranchised felon population.[4] We hypothesize that, as a result, the announcement may have motivated Virginia's elected felony prosecutors to increase charge and sentence severity for felony defendants they perceived as politically opposed, while decreasing them for those defendants they perceived as politically aligned, in order to mitigate changes to the partisan composition of the electorate.

In order to test this hypothesis, we collected all individual criminal charges filed or sentenced in Virginia Circuit Courts between 2010 and 2020. Our raw data featured 896,824 felony and 333,354 misdemeanor charges, each with detailed accompanying information on the name, race, sex, and charging district of the defendant, and on the charge's progress through the justice system from indictment to disposition and sentence. From these data we constructed a case-level dataset by aggregating all charges filed or sentenced on the same day for the same defendant, and ordering multiple charges within a single case by the severity of their maximum sentences. This process yielded 386,992 felony cases and 134,695 misdemeanor cases filed or disposed between 2010 and 2020, 56,023 of which were filed or disposed in 2016.

We analyzed the effect of McAuliffe's announcement on charging and sentencing behavior in our data using a nonparametric regression-discontinuity-in-time (RDiT) approach, then checked for robustness using both a local randomization approach and a

[3]Commonwealth of Virginia Executive Department, "Order for the Restoration of Rights," Apr. 22, 2016. https://web.archive.org/web/20160502123748/https://commonwealth.virginia.gov/media/5848/order_restoring_rights_4-22-16.pdf; and https://www.census.gov/library/visualizations/2016/comm/citizen_voting_age_population/cb16-tps18_virginia.html.

[4]See, e.g., https://www.progress-index.com/story/news/crime/2016/06/05/at-voter-rights-restoration-rally/28224272007/.

simple series of pooled differences-in-means. Our main results disaggregate defendants by political alignment with their prosecutors, then compare the charging and sentencing outcomes experienced by each group immediately *before* the rule change, to those immediately *after* it. Following Helmke, Kroeger and Paine (2022) and Eubank and Fresh (2022), we construct an alignment indicator by assuming that in light of racially polarized voting patterns (see, e.g., Kuriwaki et al., 2024; Mason, 2018), a defendant's race is informative—though it need not be determinative—of her likely political affiliation.[5] Because defendant race is noted on every criminal case file, this information is immediately and costlessly available to prosecutors.

Our results are as follows. First, immediately after the announcement, we find that defendants who were likely politically unaligned with their prosecutor experienced increases in felony charge severity and actual sentence length, while likely-aligned defendants experienced decreases. These findings persist across a range of different RDiT specifications and bandwidth choices, are broadly robust to both pooled difference-in-means regression analyses and local randomization inference, and do not appear to depend on prosecutor office size. Providing further support for our theory, we show that the announcement had no effect on charging or sentencing for the subset of (non-disenfranchising) misdemeanor cases handled in Circuit Court over the same period.

Second, these effects appear to have occurred via a range of mechanisms. Shifts in sentence severity persist when we limit our analysis to cases charged prior to the announcement, which suggests that the policy change may have directly affected prosecutors' sentencing decisions. However, the announcement also appears to have affected prosecutors' initial charging decisions and their willingness to engage in bargaining mid-case—for example, by reducing or dismissing charges. To understand the announcement's total effect on sentence severity, we estimate the pre-announcement relationship between charge severity and sentence severity and use it to predict the indirect effect

[5]Racial sorting into parties is a well-known feature of contemporary American political life (e.g., Mason, 2018) which extends to the felon population (Burch, 2012).

of post-announcement changes to charge severity on sentencing outcomes. Combining these results with the announcement's direct effect on sentencing, we find that the total effect may have been an average sentence increase of four to seven months for unaligned defendants, and an average decrease of one to three months for aligned defendants.

Legislative attempts to manipulate the composition of district-level or state electorates through gerrymandering and voting restrictions may be self-interested — each legislator intending to benefit herself — or motivated by broader partisan concerns, or both. Likewise, prosecutors may have incentives to strategically disenfranchise criminal defendants to improve their own chances of reelection, and/or to improve the chances of their favored party more broadly. With regard to *personal* incentives, so long as prosecutors can readily infer defendants' likely partisanship and are uncertain about their precise future margins of victory, manipulation may be worthwhile even when the total number of defendants affected is low. But prosecutors may also manipulate felon status for *partisan* reasons, to tip the scale in other, potentially closer electoral contests. McAuliffe's announcement seems likely to have triggered partisan feeling: his changes to Virginia's disenfranchisement law were described by Republicans as an attempt to guarantee Virginia for Democrats in the 2016 presidential election (Horwitz and Portnoy, 2016), and were accompanied by efforts to boost voter registration among eligible felons.[6] Thus, it is possible that prosecutors were especially motivated to manipulate felon status due to the perceived salience of felon voting restrictions for the 2016 presidential election.

We conduct three series of tests to disentangle the personal from the partisan motivation, and find some evidence in favor of the partisan channel. We begin with the theory

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[6] The state's online restoration of rights announcement linked to a voter registration page (See https://web.archive.org/web/20160501210020/http://commonwealth.virginia.gov:80/judicial-system/restoration-of-rights/) and post-announcement, activists held voter registration rallies for newly enfranchised felons around the state (See https://www.progress-index.com/story/news/crime/2016/06/05/at-voter-rights-restoration-rally/28224272007/.)

that prosecutors who wish to shape the outcome of an election should wait to dispose of cases for aligned defendants until after the election, but dispose of cases for unaligned defendants before it. If prosecutors engage in such behavior only around their own elections, this would be evidence for a personal motive; if prosecutors engage in this behavior more generally, it is evidence for partisan concerns. Accordingly, we examine patterns in criminal case outcomes in the weeks around state (odd-year) and federal (even-year) elections in Virginia from 2012 to 2019. We find little evidence of an effect in state election years, but do find some evidence of an effect in federal election years. This suggests that prosecutors may be motivated by broader partisan concerns.

If partisan feeling motivates prosecutors to manipulate felon status, the announcement's effect on charging and sentencing should be stronger among more enthusiastic partisans. We use prosecutors' donor histories in the Database on Ideology, Money in Politics, and Elections (Bonica, 2016) to examine how three measures of partisan political engagement — ideology, donation recency, and donation size — affected prosecutors' responses to the announcement. We find patterns consistent with donors, especially more recent donors, showing greater responsiveness to the announcement. We also find some evidence that bigger donors were more responsive.

Finally, if prosecutors are motivated by their desire to stay in office, more politically vulnerable prosecutors might be expected to have responded more strongly to the announcement. Because vulnerability is endogenous to political aptitude, this possibility is difficult to test. Nevertheless, as a first cut, we code prosecutors as politically "threatened" if their last election featured a challenger who received at least 10% of the vote. We find that political threat, as measured, has no effect on charging and sentencing behavior post-announcement.

We close by investigating whether our results could be attributable to other mechanisms, such as electoral responsiveness or prosecutors' personal support for/opposition to felon disenfranchisement. We find little evidence for these possibilities. Prosecutors never increase or decrease sentence severity against all defendants post-announcement, as would be consistent with simple electoral responsiveness arguments. Moreover, prosecutors of both parties exhibit increased punitiveness towards the unaligned and increased leniency towards the aligned. We also find no evidence that prosecutors' personal attitudes toward felon disenfranchisement shaped their behavior: the effect is similar among prosecutors who opposed the Governor's executive order and those who did not.

Our paper is relevant to the growing literature on justice officials as strategic political actors. Existing work in this field has focused on the incentives for prosecutors and judges to manipulate (and often, maximize) charges, conviction rates, or sentences as a way of signaling their attributes to voters or political principals, with a view to being reelected or promoted (e.g., Gordon and Huber, 2002, 2007, 2009; Mungan, 2017; Bandyopadhyay and McCannon, 2014; Dyke, 2007). Our contribution is to show that elected justice officials may also manipulate charges, convictions, and sentences in order to directly shape the electoral environment.

Our paper is also relevant to scholarship on the attempts by political actors to manipulate the composition of the electorate (Bateman, 2018; Helmke, Kroeger and Paine, 2022; Eubank and Fresh, 2022; Behrens, Uggen and Manza, 2003). The literature in this area has focused primarily on the incentives of legislators and elected leaders to shape the electorate in their favor: with regard to felon disenfranchisement laws, for example, scholars have raised the possibility that these laws may have been strategically imposed (e.g., Helmke, Kroeger and Paine, 2022; Eubank and Fresh, 2022), but have not considered that such laws might also be strategically *implemented*. Here, we contribute by demonstrating that the state-level imposition of such laws may motivate local-level officials to manipulate their application.

Elected Prosecutors and Felon Voting Laws

Consider an interaction between a felony defendant and a prosecutor. The prosecutor can offer a plea deal which can accepted or rejected by the defendant. Consistent with prior work among political economists and legal scholars (Landes, 1971; Grossman and Katz, 1983; Baker and Mezzetti, 2001; Gordon and Huber, 2009; Stuntz, 2011), suppose that the prosecutor has discretion to select the precise plea offer that maximizes his utility. If the defendant rejects the plea, the players go to trial, where with some probability the defendant is convicted and suffers a prison sentence, and with complementary probability she is acquitted. Trial comes at a privately known cost to the defendant, and her utility is decreasing in the length of her sentence. Trial is also costly to the prosecutor, but his utility is increasing in the length of the defendant's sentence.

In this context, the prosecutor must trade off a higher sentence in his plea offer against a higher likelihood of being rejected and going to trial. In equilibrium, he offers the plea sentence that optimally takes into account his own known cost of trial, the defendant's unknown cost of trial, and the defendant's expected sentence at trial. (In the Appendix, this argument is presented formally and the game is solved in full.)

Now suppose that there is a felon disenfranchisement rule on the books, and that the prosecutor is also concerned about the outcome of an election and suspects the defendant will vote for the opposite party. If the disenfranchisement rule is such that the defendant would be indefinitely disenfranchised after conviction, then compared to the equilibrium described above, the prosecutor adjusts his plea offer *downward* to induce the defendant to accept and lose the right to vote. By contrast, if the defendant would only be disenfranchised for the length of her sentence, the prosecutor adjusts his optimal plea offer *upward* to inflict a longer sentence. While in both cases the size of the adjustment depends on how much the prosecutor cares about preventing the defendant from voting, some shift is always optimal. If the prosecutor suspects the defendant is a copartisan, his behavior reverses: under an indefinite disenfranchisement rule he adjusts the plea offer upward (to discourage the defendant from taking it) and in a sentence-dependent regime he adjusts it downward (to minimize her disenfranchised time). If the felon disenfranchisement rule is a hybrid of these two possibilities—with some probability a defendant is indefinitely disenfranchised upon conviction, and with reciprocal probability she is disenfranchised only for her sentence—we show that an increase in the probability that a defendant's disenfranchisement is sentence-dependent corresponds to an increase in plea offer severity if the defendant is politically unaligned, and a decrease if she is aligned.

So far, we have assumed that the defendant does not weigh the cost of disenfranchisement in making her plea decision. This is empirically plausible for several reasons: defendants may not know about felony disenfranchisement rules, may discount the future loss of voting rights relative to their present concerns, or the cost of losing the franchise may be very small relative to the cost of prison. Nevertheless, in the Appendix we extend the model to consider the case where the defendant dislikes disenfranchisement. We show, inter alia, that when disenfranchisement is sentence-contingent, an aligned defendant's disutility from disenfranchisement causes the prosecutor to further adjust his plea offer downward. An unaligned defendant's disutility from disenfranchisement reduces, but does not fully counteract, the prosecutor's incentive to adjust his plea offer upward, unless the defendant's disutility from disenfranchisement is very high relative to its benefit to the prosecutor.

How might these theoretical intuitions apply in the case of Virginia? First, the Governor's executive order effectively transformed the felon disenfranchisement regime from a hybrid regime to a sentence-dependent regime: prior to the announcement, multiple factors determined disenfranchisement status after conviction; after the announcement, only sentence length determined disenfranchisement. Consequently, the importance prosecutors assigned to decreasing (increasing) the defendant's probability of voting via the imposition of longer (shorter) sentences should have increased.

Second, the executive order directly altered the composition of the electorate by immediately enfranchising formerly disenfranchised felons, and motivating widespread efforts to register them to vote. If we expect that pre-announcement, prosecutors were using a wider set of tools to disenfranchise unappealing defendants, this alteration should have resulted in a sudden increase in the share of unaligned voters in each district, which should have negatively shocked prosecutors' own electoral viability and that of their copartisans. These shocks should have heightened the importance prosecutors assigned to decreasing (increasing) the defendant's probability of voting via the imposition of longer (shorter) sentences. These two effects suggest the following hypothesis:

Hypothesis: The unexpected April 22, 2016 changes to felon disenfranchisement policy in Virginia caused the state's elected prosecutors to increase the length of the felony sentences imposed on politically unaligned defendants, and decrease the length of the felony sentences imposed on politically aligned defendants.

This hypothesis is independent of the prosecutor's precise reasons for shaping the electorate. These reasons might be personal: an office-seeking prosecutor might plausibly manipulate the electorate purely to improve his own chance of reelection. They might be broadly partisan: a policy-seeking prosecutor might plausibly manipulate the electorate in order to improve the electoral chances of copartisan politicians (whose policy positions he is likelier to approve). They might also be a mix. In a later section, we provide suggestive evidence that prosecutors' attempts to manipulate the electorate are motivated at least in part by a desire to help copartisans.

Context: Criminal Justice in Virginia

On April 22, 2016, Virginia Governor Terry McAuliffe held a press conference in which he announced an executive order implementing sweeping changes to the state's felon disenfranchisement rules.[7] Before the order, the criteria for voting rights restoration had depended on the offense, with some offenders eligible to be considered for rights restoration upon completion of their sentence provided they were not currently facing felony charges, and others required to complete a 3-year waiting period and submit a

[7]Commonwealth of Virginia Executive Department, "Order for the Restoration of Rights," Apr. 22, 2016. https://web.archive.org/web/20160502123748/https://commonwealth.virginia.gov/media/5848/order_restoring_rights_4-22-16.pdf

written application.[8] After the order, *all* felons who had completed their felony sentences of incarceration or supervised release could expect to have their right to vote immediately and automatically restored.[9] According to the Governor, an immediate result of this change was the automatic enfranchisement of 200,000 felons who had already completed their sentences—a 3.3% expansion in the statewide voter pool.[10]

For our purposes, the Virginia Governor's announcement provides an excellent opportunity to test whether elected prosecutors attempt to shape the electorate by selectively imposing disenfranchising sentences. This is for three reasons. First, the policy change was unexpected, surprising all but a small group of close political advisors and allies. State legislators, bureaucratic officials—even voter registrars and prosecutors—had been intentionally kept in the dark, to prevent them from attempting to scuttle the changes before they were announced (Zullow and Moomaw, 2016).

Second, it sharply and clearly limited how a felony conviction could result in disenfranchisement. Post-announcement, *all* felons were to be automatically enfranchised after the length of their prison or probation sentence, and any individual convicted of a felony but not sentenced to prison or probation time kept her right to vote. This allows for a clear hypothesis: in the aftermath of the announcement, prosecutors concerned with shaping the local electorate should have shifted their focus to manipulating the severity of the sentences imposed on criminal defendants.

Third, a unique feature of Virginia's criminal justice system makes it an especially useful context in which to study the strategic incentives of elected prosecutors: while in many states, both prosecutors and judges are publicly elected, in Virginia this is not so.

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[8] Archived instructions for the restoration of voting rights in Virginia as of April 17, 2016 are available here: https://web.archive.org/web/20160417134119/http://commonwealth.virginia.gov:80/judicial-system/restoration-of-rights.

[9] Misdemeanor convictions and sentences are never disenfranchising in Virginia.

[10] See https://www.census.gov/library/visualizations/2016/comm/citizen_voting_age_population/cb16-tps18_virginia.html

Instead, felony prosecutors[11] ("Commonwealth's Attorneys") are chosen in district-level competitive elections every four years—while the Circuit Court judges who handle felony cases are appointed by the legislature to eight-year terms.[12] These judicial appointments occur in an almost consociational fashion, beginning with a judge's nomination to the Courts Committee by the local Bar Association (often in conjunction with other interested groups), proceeding to unanimous election by all Senators representing a given judicial circuit, and usually ending with unanimous confirmation by the Senate.[13] This suggests that Commonwealth's Attorneys are likely, and Circuit Court judges unlikely, to respond to electoral or partisan considerations.

Data

Our raw data consisted of all Circuit Court charges filed between 2010 and the first months of 2020 in 118 of Virginia's 120 Circuit Courts.[14] The Office of the Executive Secretary at the Supreme Court of Virginia provided access to Circuit court cases, which we complemented with additional sentencing data from Virginiacourtdata.org.[15] These data include, for each charge, the date of indictment; the geographic district in which the charge was brought; the defendant's full name, race, gender, and date and month of

[11]Prosecutors' offices have discretion over whether to prosecute misdemeanors (Virginia Code §15.2-1627(B)); offices vary widely in their treatment of misdemeanor offenses.

[12]Criminal cases are distributed across District Courts and Circuit Courts in Virginia. The latter have exclusive jurisdiction over felonies.

[13]See, e.g., "A Legislator's Guide to the Judicial Selection Process," *Virginia Division of Legislative Services*, available at http://dls.virginia.gov/pubs/legisguidejudicselect.pdf

[14]We are missing Circuit Court data from the districts of Alexandria and Fairfax because these districts do not use the state case management system.

[15]This website, run by software engineer Ben Schoenfeld, compiles bulk data on Virginia state court outcomes. See https://virginiacourtdata.org/.

birth; the eventual disposition of the charge; and any sentence imposed. Most charges also include information on their movement through the system, including charge reductions and hearings.

We aggregated all criminal charges up to the case level by treating all charges resolved on the same day, for the same individual, as belonging to the same case. This yields a dataset of 521,687 cases charged or sentenced between 2010 and 2020. Among these cases, 386,992 involve felony offenses, of which 4,655 are probation violations on prior felony convictions. The remainder of the cases involve only misdemeanor charges. Since standalone misdemeanor charges are as a general rule heard in District Court, not Circuit Court, the reasons for these cases' inclusion vary significantly across cases, across districts, and across time. They include being an appeal from a District Court misdemeanor case, being ancillary to a dismissed felony charge, or, in some districts, because the Circuit Court Clerk and local prosecutors have agreed that sufficiently serious misdemeanors may be prosecuted in Circuit Court (Code of Virginia, section 19.2-190.1. "Certification of ancillary misdemeanor offenses.")

Because misdemeanors do not result in disenfranchisement, we omit these cases from our main analysis, leveraging them instead to conduct a set of placebo tests. We also exclude cases reactivated by of probation violations, for two reasons. First, active probationers were disenfranchised both before *and* after the announcement. Second, probation violations often cannot be linked back to the original felony sentences imposed, meaning that we cannot determine the extent to which the severity of probation violation punishments is a function of the severity of the original conviction. However, we show in the Appendix that the inclusion of probation cases does not materially alter our results.

Finally, we merged our criminal case records with district-level vote shares in the 2012 and 2016 presidential elections and incumbent Commonwealth's Attorneys' party membership, election years, and election outcomes (including vote shares),[16] as well as the

[16]All collected from the Virginia Department of Elections, at https://historical.elections.virginia.gov/.

partisan identities of their challengers (from Hessick and Morse, 2019). We supplemented this information with data on Commonwealth's Attorney office size from a 2018 Virginia Compensation Board survey.[17]

Measurement

Testing our theory required constructing measures of both prosecutor-defendant alignment and case-level charge and sentence severity. We operationalize prosecutor-defendant alignment as follows.

Defendant race as a proxy for political alignment As discussed above, race is indicated on every defendant's case file, and is a commonly recognized heuristic for partisan affiliation. In Virginia, where the vast majority of citizens report their race as white or African-American, this heuristic appears accurate: 76% of African-Americans statewide described themselves as Democrats or leaning-Democrat in 2014, as compared to 27% of white Virginians (Pew Research Center, 2014). Race appears to be a similarly strong predictor of political affiliation among southern felon populations: in North Carolina and Florida, more than 80% of registered African-American felons, but only around 33% of registered white felons, were registered Democrats (Burch, 2012, 10) . We leverage this relationship with the assumption that prosecutors in all districts view African-American felony defendants as more likely to vote for Democrats, and white felony defendants as more likely to vote for Republicans. While this assumption does not take into account cross-district variation in racial political preferences, and particularly in white support for Democrats, we believe it is appropriate for two reasons.

First, there is methodological value in using a simple alignment rule applied equally to all prosecutors as our main specification. While it is possible that prosecutors make

[17] "Workgroup Study of the Impact of Body Worn Cameras on Workload in Commonwealth's Attorneys' Offices," available at https://www.scb.virginia.gov/docs/bodycameraworkgroupreport.pdf.

more nuanced inferences about defendants' political preferences, these nuances are hard to account for in a principled manner, given the limitations of our data. Second, to the extent that white support for Democrats varies across districts, we argue that even in districts where a randomly selected white *person* would be likelier than not to lean Democratic, this is not the case for a randomly selected white felony defendant, because white support for Democrats is tied to other demographic indicators which are sparse in the felony defendant population. Specifically, support for Democrats tends to be higher among white women than men, and to increase with education,[18] while white felony defendants are heavily male and tend to have low levels of formal education.[19] Additionally, there is some evidence that Democratic party operatives make similar assumptions about white felons' political preferences: during a dead heat governor's race, Washington State's Democratic party chairman argued that illegal felon voting had tilted the race toward the Republican candidate, because the felon population was "nonunion, blue-collar, male [and] Caucasian ... the prime demographic Republicans target" (Postman, 2005).

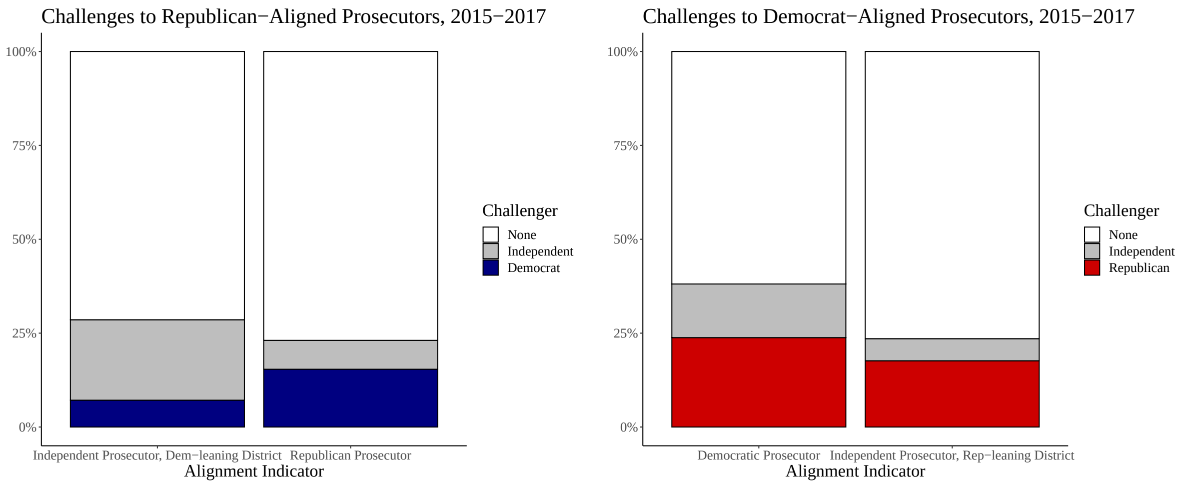
Prosecutor political party and alignment Given this assumption, when prosecutors campaign explicitly as Republicans or Democrats, we treat a prosecutor-defendant pair as politically aligned when their parties match. Thus, for example, a white defendant and a Republican prosecutor are treated as politically aligned. However, approximately half of Virginia Commonwealth's Attorneys run for election as Independents. Because

[18] In Virginia's D.C.-adjacent Democratic strongholds, an overwhelming majority of whites — 65% to 85% — hold a B.A. degree or higher. See the county-level Virginia education map presented by the National Institute on Minority Health and Health Disparities, available at https://hdpulse.nimhd.nih.gov/data-portal/social/map?.

[19] According to one study, only 5% of white males formerly incarcerated in a state prison held a BA degree or higher. See Couloute, Lucius (2018). "Getting Back on Course: Educational exclusion and attainment among formerly incarcerated people." *Prison Policy Initiative.* Available at https://www.prisonpolicy.org/reports/education.html#table3.

Virginia does not have an Independent political party, and Independents are barred under state law from receiving endorsements or funding from the Republican and Democratic parties,[20] we cannot determine an independent prosecutor's likely voter base from party endorsements or donations. Moreover, voters in Virginia who identify as Independent are evenly split between those who lean Democrat and those who lean Republican (Pew Research Center, 2014), suggesting that the label has no inherent partisan association.

Figure 1: Challenger Party by Theorized Alignment Status



Left panel: party identity of candidates who challenged Independent prosecutors in Democratic-leaning districts, compared to the party identity of candidates who challenged Republican prosecutors. Right panel: party identity of candidates who challenged Independent prosecutors in Republican-leaning districts, compared to those who challenged Democratic prosecutors. We categorize a district as Republican-leaning or Democrat-leaning based on which presidential candidate won the majority in that district in the 2012 presidential election.[21]

Here, we begin with the premise that voters view a politician's party membership as informative of her ideology and policy goals. If this is true, then because Independents are barred from entering into alliances with Virginia's two recognized political parties, in a district with a partisan lean, an incumbent Independent should expect her partisan chal-

[20]Virginia Department of Elections. "How to Run for Local Office." June and November 2021 Local Candidates Bulletin. Available at https://www.elections.virginia.gov/media/candidatesandpacs/2021-11-02_Gen_Bulletin_Local_Offices_rev_12-16-2020.pdf.

lengers to hail from the political party with greater voter support in the district. Figure 1, which presents challenges to incumbent prosecutors 2015-2017 by party affiliation (and for Independents, by district partisan lean), shows support for this premise. Independent incumbent prosecutors in blue-leaning districts were *only* challenged by Democrats and other Independents, while those in red-leaning districts were *only* challenged by Republicans and other Independents.

If voters view party membership as informative, an Independent incumbent facing a partisan challenger should not expect the votes of citizens who are dedicated supporters of the challenger's party. By the same logic, she *should* expect the votes of citizens who are dedicated members of the other party, because these citizens will prefer her to her partisan opponent. These calculations suggest that an Independent incumbent should in terms of substantive policy position herself in the ideological center, close enough to the dominant party to build a majority coalition—but in terms of electoral strategy, she should treat likely minority-party supporters as ' 'aligned,' and likely majority party supporters as 'unaligned.'

Table 1 presents our overall operationalization of political alignment among prosecutor-defendant pairs.

Table 1: Prosecutor-voter alignment

	White felon	Black felon
Republican	aligned	unaligned
Democrat	unaligned	aligned
Independent, Republican county	unaligned	aligned
Independent, Democratic county	aligned	unaligned

Charge and Sentence Severity Because prosecutors may influence the length of a felony sentence directly, via their choice of recommended sentence, or indirectly, via the charges they file, we constructed measures of both charge and sentence severity. To measure charge severity, we used Virginia's sentencing guidelines to determine the maximum permissible sentence for each charge in a case, then coded the case's charge severity as the maximum sentence attached to the most severe ('top') charge filed. Nearly all permissible maximum sentences in almost all felony cases in our data take on one of five distinct values: five, ten, twenty, or forty years in prison or life imprisonment. A small number of charges take on other values, including three years, thirty years, and fifty years. This measure of charge severity is a conservative estimate, because it does not account for the severity of *all* charges in a given case, but only the top charge. Because we cannot be certain about the actual length of a 'life' sentence, we recode these sentences as fifty-year sentences.

To measure sentence severity, we simply calculated the total *felony* prison sentence assigned after conviction in each case. Where multiple charges are sentenced in the same case and the sentences were set to run consecutively, we aggregated the charge-level data by summing each count's sentence length in days. Where sentences were set to run concurrently, we use the longest sentence imposed. This measure is likewise a conservative, *minimum* estimate of actual sentence severity, since it does not include supervised probation time. We exclude probation time for two reasons. First, many probation sentences in the data are 'indefinite.' These sentences feature neither a maximum nor a minimum time: they end when the court actors involved in the case decide they should. Second, even definite terms of probation are indefinite in practice because infractions during the probation period can be used (at the prosecutor's discretion) to reset the probation clock.[22]

Empirical Strategy

Treating McAuliffe's announcement as an unexpected shock to felon disenfranchisement rules in Virginia, we use several analytical methods to test for discontinuous, alignment-specific shifts in prosecutorial punitiveness in the weeks immediately after the shock.

[22]See Code of Virginia § 19.2-304, see also Sentencing Revocation Report and Probation Violation Guidelines (2016) at http://www.vcsc.virginia.gov/worksheets_2015/SRR_booklet2015revised.pdf.

This short-term focus, together with the surprise nature of the announcement, allows us to minimize confounding by unrelated, longer-term patterns in the composition and treatment of criminal cases.

First, taking advantage of our daily district-level data, we use a local-linear regression discontinuity in time (RDiT) approach to test for the effect of the governor's announcement on charge and sentencing outcomes under the assumption that absent the announcement, potential outcomes would have been a smooth function of time. Our data are well-suited to this approach, which permits seasonality in criminal outcomes so long as these outcomes change smoothly in time.[23] However, to address any inferential shortcomings in our RDiT analysis, in addition to a set of robustness and placebo tests we also present results from a simple difference-in-means fixed effects regression analysis, which relies on the assumption that the announcement was as-if random. In the Appendix, we also present results from a local randomization analysis, suggested by Cattaneo, Idrobo and Titiunik (2024) as a robustness check for data with discrete running variables.

The following local-linear regression model is our main RDiT specification throughout:

$$y_{ijt} = \beta_0 + \beta_1 \mathbb{1}(t > t_{ann.}) + f(t - t_{ann.}) + \alpha_j + X_{it} + \epsilon_{ijt}. \tag{1}$$

Here, y_{ijt} is a measure of punishment severity, for example, the total prison sentence for case i in district j at time t, $t_{ann.}$ is the date of the announcement, $f(\cdot)$ is a smooth function of distance from the same, and the parameter of interest, β_1, captures the discontinuous shift in punishment severity following the announcement. In most specifications, we include district fixed-effects (α_j) to account for cross-district differences in sentencing

[23] While data lacking cross sectional variation and with observations at few running variable values can be problematic for RDiT designs, especially if the treatment is not quasi-randomly assigned (see, e.g., Hausman and Rapson, 2018), our treatment *is* plausibly quasi-random, and our data contain cases charged and decided every day across numerous districts, suggesting that local-linear RD estimation remains appropriate.

norms, crime, constituency preferences, and prosecutor's office (each district elects one chief felony prosecutor). In some specifications, we also include pre-treatment dummies X_{it}, which capture a defendant's gender and whether the type of crime prosecuted was a violent crime, a drug crime, a property crime, a traffic crime, or some other infraction. Our theoretical prediction is that the sign of the parameter of interest, β_1, should be negative for defendants aligned with their prosecutors, and positive for unaligned defendants. To test this prediction, we estimate Equation (1) separately according to the felon's political alignment vis-a-vis the prosecutor.[24] Throughout, we use MSE-optimal bandwidths with the bias-correction method proposed by Calonico, Cattaneo and Titiunik (2014), and standard errors clustered at the district-level.[25]

After reporting our main regression results, we turn to an investigation of the announcement's total effect on sentencing: i.e., its effect on both direct changes to sentence severity and indirect changes to sentence severity through changes in charge severity. We test for a *direct* sentencing effect by estimating the announcement's effect on sentence severity only for cases charged prior to the announcement. We also examine how prosecutor use of specific sentencing tools shifted post-announcement.

Testing for an *indirect* sentencing effect via changes in charge severity is harder, because cases charged after the announcement are necessarily sentenced after it as well,

[24]Note that our theory implies that aligned and unaligned defendants are not comparable before *or* after McAuliffe's announcement: while the announcement should have caused a *change* in prosecutors' respective treatment of these two classes of defendant, they should have been treated differently before the announcement as well.

[25]Kolesár and Rothe (2018) and others have shown that clustering on the running variable is inadvisable, even when it is discrete. We therefore first estimate local-linear regression discontinuity results, clustering by district and relying on the large number of unique values in our (daily) running variable, then probe the robustness of our results by conducting additional difference-in-means tests as well as local randomization inference in the neighborhood of the threshold (see Cattaneo, Idrobo and Titiunik, 2024).

making it difficult to distinguish the effect of shifts in charge severity from direct sentence manipulation. Our solution is to estimate the pre-announcement relationship between charge severity and eventual sentence length. We then use this relationship to derive bounds on the size of the announcement's effect on sentence severity, through changes in charge severity.

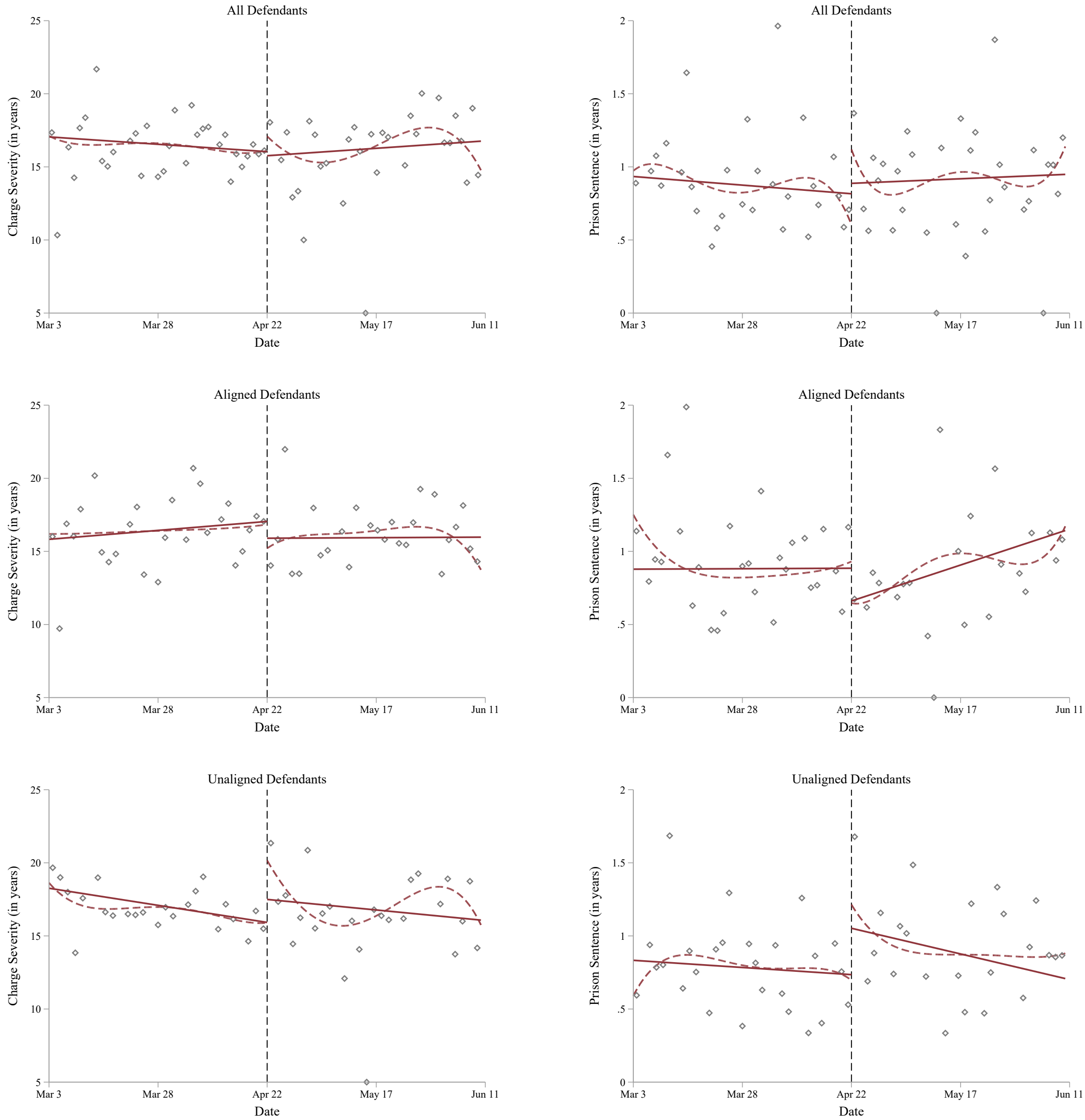
We conduct a number of robustness and placebo tests to bolster confidence in our results. In later sections, we conduct additional tests to disentangle our explanation from one based on districts' or prosecutors' ideological preferences.

Main Results

We begin with a graphical assessment of the announcement's effect on felony charge severity and felony sentence length. The results are presented in Figure 2. The left-hand column depicts binned averages of the severity of the top charges brought against defendants in Circuit Court in a twelve-week window around the announcement, adjusted for prosecutor fixed effects. The right-hand column depicts binned averages of the lengths of the actual felony prison sentences imposed on defendants in Circuit Court during the same time period, adjusted for prosecutor fixed effects. We separately fit linear predictions and kernel-weighted polynomial curves to the data on either side of the announcement. The top row presents the announcement's effect on *all* charges and sentences. The middle and bottom rows present the announcement's effect on charges and sentences for aligned and unaligned defendants, respectively. Consistent with our theory, we observe significant post-announcement shifts by alignment status: sharp increases in average charge and sentence severity for unaligned defendants, and sharp decreases for aligned defendants.

Local-Linear Estimates Table 2 presents local-linear estimates of the immediate effect of McAuliffe's order on the average severity of the top charges filed (Panel A) and prison sentences imposed (Panel B) in Circuit Court, disaggregated by alignment status. For each defendant type, the first column presents unadjusted results for comparison.

Figure 2: Announcement Effect on Charge (left column) and Sentence Severity (right column)



The top row in the figure displays the unconditional, within-district effect of the announcement on average charge severity (left) and prison sentences (right) before and after the announcement. The next two rows disaggregate these results for aligned and unaligned defendants, respectively. The dashed line represents a triangular kernel-weighted polynomial fit. The solid line represents a linear fit.

Table 2: Effect of McAuliffe's Order, By Alignment Status

	Unaligned Defendants			Aligned Defendants		
	Panel A. Charge Severity					
RD Estimate	2.945	3.049	2.464	-2.108	-2.174	-1.787
	(1.534)	(1.205)	(1.184)	(1.77	(1.065)	(1.076)
Left Intercept	16.771	16.78	16.694	18.581	18.754	18.805
	(0.983)	(0.716)	(0.741)	(1.349	(0.874)	(0.846)
Bandwidth	67.053	61.664	51.763	70.306	71.345	57.194
Eff. Observations	6161	5647	4699	8016	8085	6593
	Panel B. Sentence Severity					
RD Estimate	0.378	0.469	0.402	-0.17	-0.169	-0.132
	(0.358)	(0.271)	(0.273)	(0.161	(0.19)	(0.184)
Left Intercept	0.748	0.759	0.759	0.876	0.864	0.87
	(0.097)	(0.084)	(0.087)	(0.12	(0.152)	(0.137)
Bandwidth	52.6	55.082	54.025	70.994	71.84	73.568
Eff. Observations	4525	4815	4728	7802	7880	8057
District FE	No	Yes	Yes	No	Yes	Yes
Covariates	No	No	Yes	No	No	Yes

Charge and sentence severity shown in years. Left Intercept is the estimated outcome immediately before the announcement. For each defendant type, second and third columns include district-fixed effects, and third columns condition on gender and crime type. Standard errors clustered at the district-level. Only African-American and white defendants (98% of all defendants) included.

The second column adds district fixed effects to take into account variation across districts in crime rate, crime type, demographics, and prosecutor identity. The third column presents results further adjusted for defendant gender, and crime type.[26]

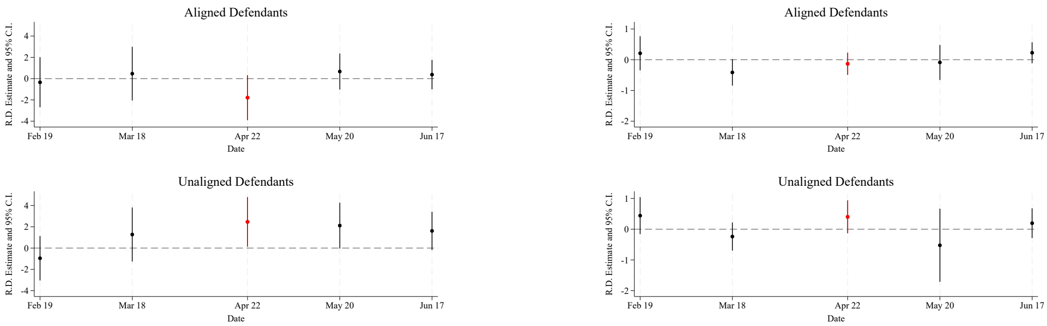
Consistent with the graphical analysis above, we find that across specifications, the announcement drove increases in charge severity and sentence length for unaligned defendants, and decreases for aligned defendants.[27] Overall, for unaligned defendants, the

[26]In our main specifications, we do not adjust for specific charges or (for sentencing) charge severity, because these are strategic choices by the prosecutor.

[27]While the results which do not include district fixed effects are appropriately signed, they are consistently weaker; this is unsurprising, given substantial demographic variation

executive order corresponded to an average increase in initial charge severity of 2 to 3 years, and an average increase in actual sentence length of 4 to 5 months. Aligned defendants appear to have experienced about a 2 year decrease in initial charge severity, and around a 1.5 month decrease in sentence severity, although the last effect is consistently statistically insignificant at conventional levels.[28]

Figure 3: Replication of Main Results, Placebo Dates



Left: Charge severity. Right: Sentence severity. RDiT estimates presented for placebo dates on the third Friday of the month, one and two months before and after the announcement. All results include district fixed effects and adjust for covariates.

These results are broadly robust to a range of alternative specifications, placebo tests, and bandwidth choices. First, in Figure 3, we replicate our analysis but replace the announcement date (April 22, the third Friday in April) with placebo dates on the third Friday of the two months prior to and following the announcement, respectively. The results are encouraging.[29] Second, if our hypothesis is correct, *only* felony charges and

across districts as well as variation in criminal activity and prosecutor behavior.

[28]Notice from the left-hand intercepts in Table 2 that aligned defendants' baseline charges/sentences were more severe than unaligned defendants,' which is consistent with prosecutors requiring greater seriousness to prosecute them at all.

[29]An informative yearly placebo analysis is not possible due both to Governor McAuliffe's prior revisions of felon disenfranchisement criteria in the spring of 2013, 2014, and 2015—although these earlier revisions were much less broad in scope and were not a "surprise"—and due to the fact that primary elections in Virginia take place in June of odd-numbered years.

sentences should have been affected by the announcement, because only felony sentences are disenfranchising. Accordingly, in Table 3, we estimate the announcement's effect on misdemeanor offenses charged or sentenced in Circuit Court in 2016. Consistent with our theory, the resulting estimates are highly insignificant across all specifications. Third, in Figure 4 below, we present the coefficient estimates for our covariate-adjusted local-linear regressions beginning with a 5-day, user-set bandwidth on either side of the announcement and increasing in two-day increments to a 90-day bandwidth. As is clear from the figure, our results track those presented in Table 2.

Table 3: Effect of McAuliffe's Order, By Alignment Status, On Misdemeanor Offenses

	Unaligned Defendants		Aligned Defendants		Unaligned Defendants		Aligned Defendants	
	Panel A. Charge Severity				*Panel B. Sentence Severity*			
RD Estimate	0.029	0.025	0.038	0.0414	-0.001	-0.006	0.016	0.016
	(0.034)	(0.031)	(0.038)	(0.034)	(0.018)	(0.019)	(0.019)	(0.019)
Left Intercept	0.909	0.910	0.900	0.899	0.035	0.035	0.036	0.036
	(0.017)	(0.015)	(0.027)	(0.025)	(0.014)	(0.015)	(0.010)	(0.010)
Bandwidth	59.074	68.279	47.580	58.100	67.161	67.354	75.832	74.064
Eff. Observations	1601	1881	1616	1675	1829	1829	2604	2584
District FE	Y	Y	Y	Y	Y	Y	Y	Y
Covariates	N	Y	N	Y	N	Y	N	Y

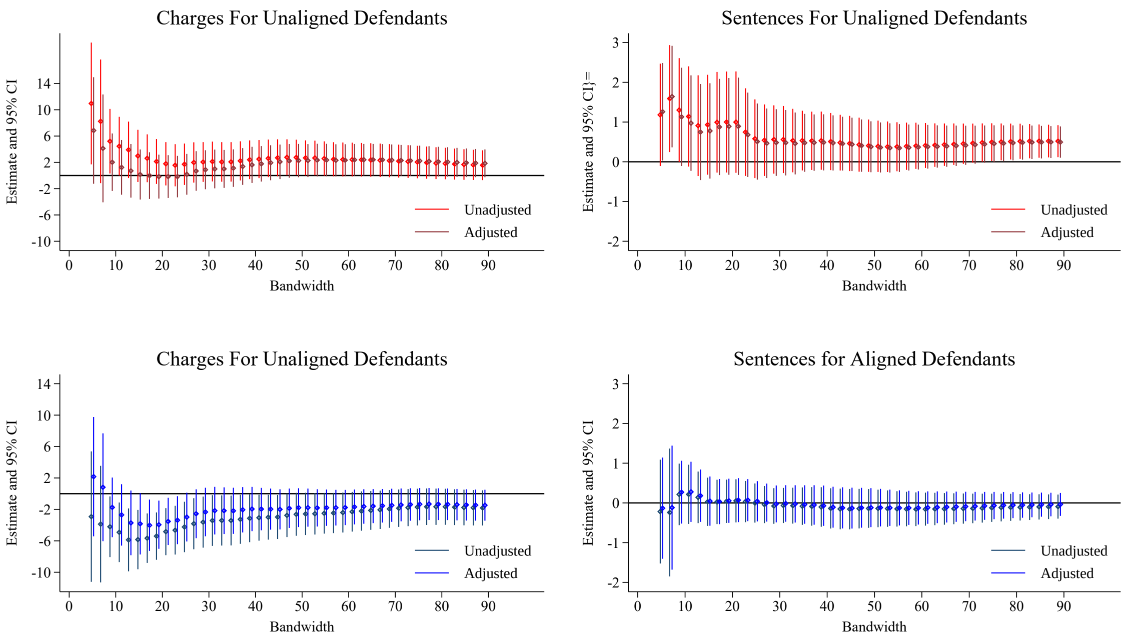
Charge and sentence severity shown in years. Left Intercept is the estimated outcome immediately before the announcement. For each defendant type, first columns include district-fixed effects, and second columns also condition on gender and crime type. Robust standard errors clustered at the district level. Only African-American and white defendants included (98% of all defendants).

In the Appendix, we conduct a number of additional tests to determine the robustness of our results to different theories of alignment and the inclusion of additional data. In Table A1, we re-estimate our main models after removing all districts headed by Independent prosecutors, and show that the results persist, although they are weaker.[30] In

[30]In this section, we also show that the results presented in Figures 5 and 6 are robust to removing Independent prosecutors.

Table A2, we test the robustness of our theory of alignment to an alternative treatment in which both white and African-American felony defendants living in strongly Democratic districts are assumed to be aligned with Democratic prosecutors. We operationalize this alternative treatment by re-coding white felony defendants as aligned, rather than unaligned, if they are charged in a district with a Democratic prosecutor in which at least 70% of the district population voted for Barack Obama in 2012. The results are consistently signed and qualitatively similar, but they are weaker. Next, in Table A3, we re-estimate our main models including probation violations, and show that the results are similar.

Figure 4: Replication of Main Results, Variation of Bandwidth



Point estimates and 95% robust confidence intervals for user-specified bandwidths beginning at 2 days and increasing to 90 in increments of two. All results include district fixed effects; "adjusted" results include pretreatment covariates.

Finally, because elected chief prosecutors generally manage a team of assistant prosecutors, we explored the possibility that the announcement's effect might depend on chief prosecutors' control over their agents: in larger districts, subordinate prosecutors might have more independence, and the announcement might have had less of an effect.

However, Figure A4 in the Appendix shows little evidence in favor of a size-based effect.

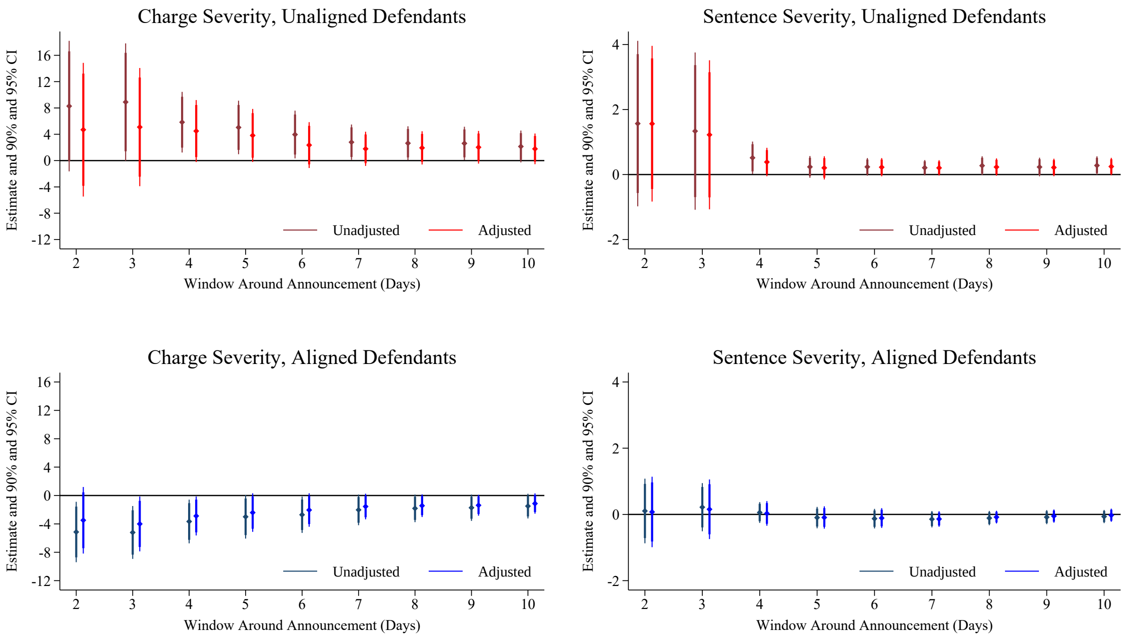
Alternative Specifications The RDiT analysis above relied on the assumption that the relevant potential outcomes for aligned and unaligned defendants are a smooth function of the running variable. Because the announcement was unexpected, an alternative assumption is that for a short enough time window around the announcement, the treatment (the announcement) was as-if randomly assigned. Accordingly, we conduct a series of simple, pooled fixed effects regressions for two-to-ten day windows around the announcement. In these estimates, the treatment takes the form of a simple indicator variable which switches from zero to one on the date of the announcement. The results are presented in Figure 5. Despite the narrow time windows, we observe consistent and almost-immediate post-announcement increases in charge and sentence severity for unaligned defendants, and consistent decreases for the aligned. These effects converge to effects similar in size to those in Table 2.

In the Appendix, we also present the results of a complete randomization inference procedure proposed by Cattaneo, Idrobo and Titiunik (2024). The procedure tests a sharp null hypothesis of no treatment for any unit for a series of small windows around the cut-point by selecting a random subset of possible treatment profiles, and measuring the extremeness of the observed test statistic against those generated from the randomly chosen profiles. We limit our examination to windows (mass points) of one to five days around the announcement, among the subset of districts for which data is available in these windows before and after the treatment. The results are shown in Table A4. Coefficient estimates are consistently signed, although they are statistically significant only for charge severity.

Mechanisms

In this section, we study the mechanisms behind our main results, asking both how, and why, these results came about. We begin with "how," working to identify the chan-

Figure 5: Effect of Announcement, Pooled Difference-in-Means Regressions



Point estimates and 90/95% confidence intervals for 2-10 day windows around the announcement. All estimates adjusted for district fixed effects; 'adjusted' results adjusted for pretreatment covariates. Standard errors clustered at the district. Clusters for charge severity: 54-96 for unaligned; 54-94 for aligned. Sentence severity clusters: 54-97 for aligned; 62-98 for unaligned.

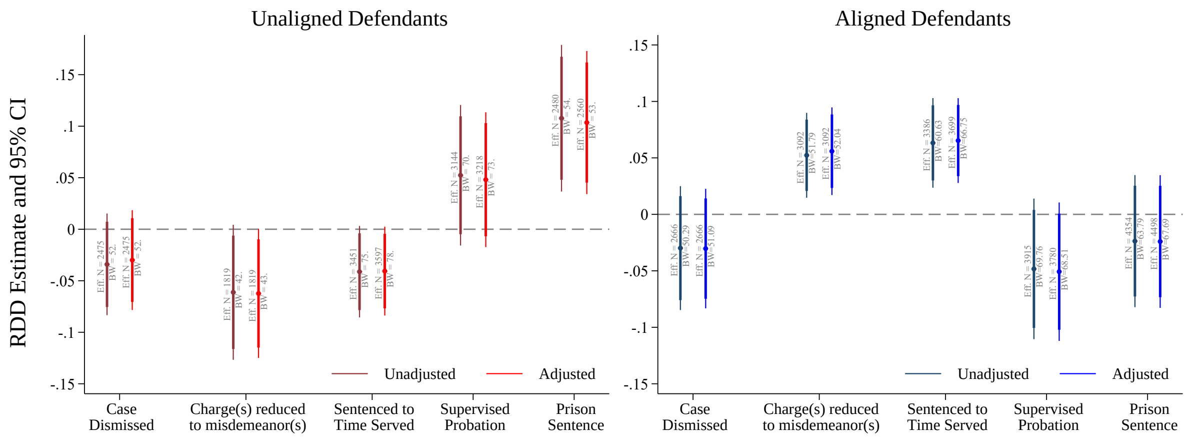
nels through which the announcement affected disenfranchisement status. Here, we ask whether prosecutors primarily adjusted charging decisions in response to the announcement or whether they directly altered sentence outcomes as well—and if so, by what methods.

Next, we consider "why." Here, we focus on, and attempt to disentangle, two possible motives for charge and sentence manipulation: a personal, office-seeking motive, in which prosecutors seek to maximize the likelihood of their own reelections, and a partisan, policy-seeking motive, in which prosecutors seek to improve the likelihood of reelection for copartisans at other levels of government.

Direct and Indirect Effects on Sentencing

We argued above that McAuliffe's announcement caused a decrease in the severity of charges and (to a lesser degree) sentences imposed on aligned defendants, and caused an increase in charge and sentence severity for unaligned defendants. We now consider the channels through which it did so. An initial question is whether the announcement had an independent effect on sentences, or whether all the effects we observed on post-announcement sentences occurred via a change in charging behavior. We adjudicate this question in two ways. First, in Appendix Table A5, we re-estimate the announcement's effect on average prison sentences, restricting our analysis to cases charged before the announcement. In some specifications, we also condition on the maximum prison sentence for the top charge in the case in an attempt to hold fixed charge severity, although as we have argued earlier, charge severity is likely endogenous to alignment before as well as after the announcement. The results are largely similar to those in Table 2, suggesting that our main sentencing results do capture some direct effect on sentencing. However, the effect is attenuated by the introduction of charge severity.

Figure 6: Effect of Announcement on Sentencing Outcomes



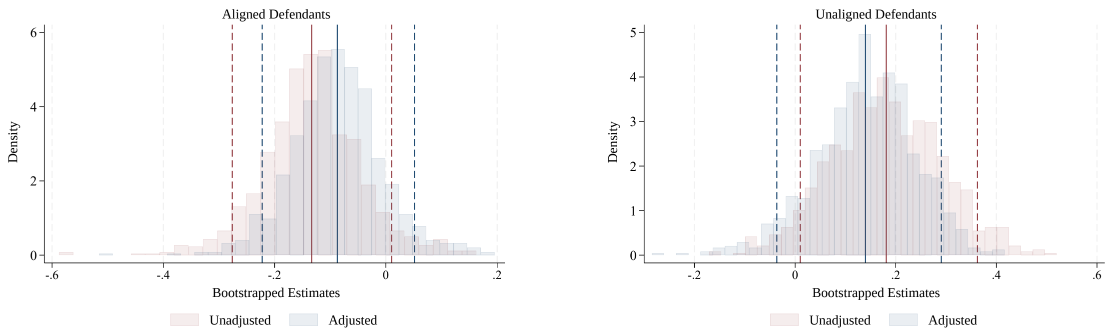
Panels provide RD estimates and 90/95% robust confidence intervals for the post-announcement change in the probability that unaligned and aligned defendants' charges are (from left to right) dismissed, reduced from felonies to misdemeanors, sentenced to time served, sentenced to supervised probation, or sentenced to prison. All estimates adjust for district fixed effects; "adjusted" estimates also control for pre-treatment covariates.

Second, we examine the announcement's effect on prosecutors' use of five tools which directly shape sentencing outcomes: outright dismissals, charge reductions to misdemeanors, and requests for time served, supervised probation, or prison time. Ideally, we would examine how prosecutors' use of each tool changed in real time after the announcement. Unfortunately, we lack dates for both informal plea agreements and amendments prior to cases' final resolutions. As a next-best strategy, we conduct a local-linear RD estimation of the announcement's effect on the incidence of each outcome (recorded at the conclusion of the case). The results, displayed in Figure 6, show that for unaligned defendants, the announcement decreased the probability that felony charges were dismissed, reduced to misdemeanors, or sentenced to time served, and increased the probability of receiving a prison or probation sentence. For aligned defendants, the announcement somewhat decreased the probability of receiving a prison or probation sentence, and significantly increased the probability of a charge reduction to a (non-disenfranchising) misdemeanor, and of being sentenced to time served. As we show in Appendix Figure A3, these effects largely persist in a series of pooled fixed effects regressions for windows of two to ten days around the announcement.

The varying strength of these different results is consistent with a situation in which prosecutors have greater short-term ability to revise sentences *downward* than *upward* — possibly because case outcomes are often the result of a negotiation between a prosecutor and defense attorney, and upward deviations from a negotiated plea deal would not be well received by opposing counsel. We find an especially strong effect on the provision of time served sentences to aligned defendants, perhaps because these sentences permit prosecutors to convict defendants of felony offenses without—after the announcement—ever disenfranchising them. In Appendix Table A6, we show that this result is robust to the Fisher complete randomization test we conducted in the previous section.

The Total Sentencing Effect The results above suggest that to understand the announcement's full effect on sentencing (and disenfranchisement) outcomes, we must separate its direct effect via sentencing strategies and its indirect effect via charges. The latter is difficult, because all cases charged post-announcement were necessarily sentenced post-announcement. To gain some purchase on this question, we estimate the pre-announcement relationship between charge severity and sentence severity, then use these estimates to predict the effect of post-announcement charging shifts, accounting for uncertainty in both predictions by performing these calculations for each of 1000 cluster-bootstrapped samples, clustering at the district level.

Figure 7: Bootstrapped Estimates of Indirect Sentencing Effect



Histograms depict bootstrapped estimates of the discontinuous shift in sentence lengths attributable to the post-announcement shift in charge severity, for aligned (left) and unaligned (right) defendants. The solid vertical lines depict averages, while the dashed vertical lines represent the 5^{th} and 95^{th} percentiles in the distributions of both unadjusted (red) and adjusted (blue) estimates.

Figure 7 displays our estimates of the indirect change in sentence severity induced by post-announcement changes in charge severity, across bootstrapped samples, for aligned defendants (left) and unaligned defendants (right). Among our samples of aligned defendants, we estimate that post-announcement decreases in charge severity led to a 30-40 day decrease in average prison sentences; although the effect skirts conventional levels of statistical significance. For unaligned defendants, we estimate that post-announcement increases in charge severity led to a 50-60 day average increase in sentence length.

Inferring the announcement's total effect on sentencing via our estimates of its direct and indirect effects requires us to assume that the mapping from charge to sentence severity was not too altered by the announcement. Given this assumption, we can place rough bounds on the size of the announcement's effect. Recall that our estimate of the direct effect on sentencing, presented in Table 3, was a prison time increase of 4 to 5 months for unaligned defendants, and a (statistically insignificant) decrease of approximately one to one and a half months for aligned defendants. If charge severity shifts substitute partially or entirely for sentencing shifts, these changes may approximate the total effect. If charging and sentencing adjustments are complementary to one another, then after the announcement, unaligned defendants may have received sentences as much as seven months longer, and the aligned as much as three months less.

Personal or Partisan Motives?

Were Virginia's chief prosecutors manipulating disenfranchisement status to further their personal electoral prospects, or to improve the electoral fortunes of copartisans more broadly? Both motives are theoretically possible. With regard to prosecutors' own races, the negligible cost of manipulation — since defendant partisanship can be readily inferred — may make it worthwhile even when disenfranchisement is unlikely to tip the scale: after all, prosecutors do not know the margin of victory in future elections, and a prosecutor who prevents a single vote for his opponent has effectively voted twice. At the same time, Virginia is known as a swing state where statewide (and federal) races are sometimes decided with very small margins, making felon disenfranchisement a potentially valuable tool in shaping outcomes statewide or even nationally—and the possibility that felon enfranchisement in Virginia might tip the scale in the 2016 presidential election was explicitly raised post-announcement (e.g., Horwitz and Portnoy, 2016).

Because the tangible result of either motive is the same in terms of prosecutor behavior, the two are difficult to disentangle. We begin with the intuition that prosecutors wishing to affect the outcome of an election might strategically front-load disposition dates for unaligned defendants *before* the election, and hold off on convicting and sentencing aligned defendants until *after* the election. This would generate a post-election *decrease* in the number of daily case dispositions for unaligned defendants, and a post-election *increase* in dispositions for aligned defendants. Prosecutors concerned only about their own electoral fortunes should arguably engage in such behavior only around their own elections, while prosecutors more broadly interested in electoral outcomes might be expected to manipulate disposition dates around every election.

To explore this hypothesis, we plotted a simple district-weighted count of felony cases decided for 50 days on either side of each Virginia election between 2012 and 2019, and separately fitted linear predictions and kernel-weighted polynomial curves to either side of the election date. The results for federal elections (even years) and state elections (odd years) are presented in Appendix Figures A8 and Figure A9, respectively. We find little evidence for an effect in state election years, but in three of four federal elections we observe higher levels of unaligned cases decided in the weeks prior to the election, followed by a substantial post-election drop.[31]

Further investigation of the partisan motive requires measures of prosecutors' relative political engagement. Because Commonwealth's Attorneys are elected, they all arguably meet baseline levels of political engagement. To determine whether prosecutors exhibited varying levels of concern for political outcomes besides their own electoral contests, we examine their donation behavior. To do so, we matched our prosecutors by first and last name (and middle initial) to Virginia donors contained in the Database on Ideology, Money in Politics, and Elections (DIME) (Bonica, 2016, 2014). The DIME database contains biennial information on individual contributions to local, state and federal elections from 1980-2022 as well as common-space ideal-point scores ("cf scores") for donors. We successfully obtained 95 prosecutor-DIME donor matches, nearly all of which we validated against occupation ("prosecutor," "Commonwealth's Attorney") employer information ("County"), and/or addresses. We cross-checked unmatched prosecutors against the OpenSecrets database and the Virginia Public Access ProjectVAData (VPAP) records

[31]It is possible that prosecutors may also set sentences to end just before or after elections, but this is not possible for us to test (and likely difficult for them to do) because we do not know how much time defendants spent incarcerated before sentencing, and this time counts against their total sentence and determines when they are released.

to confirm that they had indeed made no (recorded) contributions.

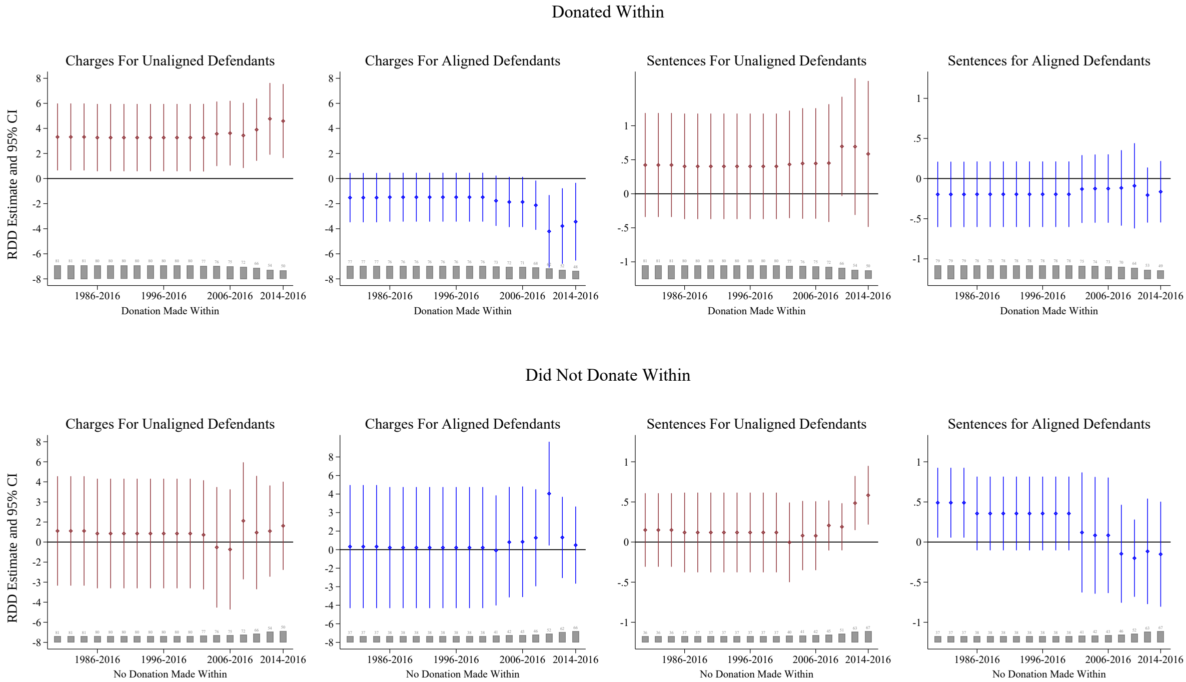
It is plausible that donor prosecutors were more invested in political outcomes than non-donors, and than those who made donations closer in time to the announcement were more invested in political outcomes at that time than those who did not. Accordingly, we begin by using DIME's biennial donation information to construct a set of measures of donation recency where the largest measure includes all donations within the entire pre-announcement interval for which we have data (1980-2016), and the smallest includes only donations made between 2014 and 2016. We then separately estimate the announcement's effect on prosecutors who did/did not donate within each interval. Figure 8 shows the district- and covariate- adjusted effect of the announcement on charging and sentencing behavior among prosecutors who did (top row) and did not (bottom row) donate within each interval. As the Figure shows, donors appear more responsive to the announcement than non-donors overall, but the overall disparities in behavior between donors and non-donors seem to increase as donation intervals become smaller, suggesting that the more recent donors may be more determinative.

Other possible measures of political engagement are ideological extremism (more extreme prosecutors may care more about electoral outcomes (e.g., Palfrey and Poole, 1987)) and donation size (more generous donors may care more about outcomes). Using the DIME dataset, we construct measures of both. Unfortunately, due to data limitations, both measures are flawed. Our measure of ideology uses common-space DIME scores (cf scores), which exist only for the subset of prosecutors who were donors. Among this subset, we treat prosecutors as "extreme" if their cf-scores exceeded the median score among prosecutors on "their" side of 0, and "moderate" otherwise.[32]

Regarding donation size, an ideal measure would adjust for characteristics like income/wealth and age to avoid inapposite comparisons between wealthy and poor or young and old donors. Because we lack information on these characteristics, we risk conflating

[32]Thus, prosecutors to the left of 0 were labeled extreme if their scores exceeded the median among all prosecutors to the left of 0.

Figure 8: Announcement Effect By Recency of Last Donation



Point estimates and 95% robust confidence intervals. All estimates adjusted for district fixed effects, crime type, and gender. Estimates unadjusted for crime time and gender are in Appendix Table A5. Clusters vary across specifications, and are depicted by gray histograms at the bottom of each figure.

them with political involvement. As a first cut, we used news reports, professional networking sites, and historical election records to hand-code prosecutors' years of tenure as Commonwealth's Attorneys, and calculated prosecutors' average annual donations during their time in office. We then treat prosecutors as large donors if their average yearly gifts while in office exceed the median.

The results are shown in Appendix Figure A6. We find no consistent differences in the behavior of 'extreme' as opposed to 'moderate' prosecutors, as measured by DIME cf scores. We do find some differences in the post-announcement behavior of larger, as opposed to smaller, donors: overall, individuals who gave larger average amounts during the course of their tenure as Commonwealth's Attorneys may have responded somewhat more strongly to the announcement. However, these results should be interpreted with caution, because the measures we use are flawed.

While the partisan motive can be proxied for empirically through other measures of political engagement, it is difficult to measure — and thus to conclusively prove or disprove — the strength of a prosecutor's personal desire for reelection. However, it seems plausible that politically vulnerable, office-seeking prosecutors might be especially prone to voter pool manipulation. This possibility is difficult to test rigorously, because political vulnerability is endogenous to political aptitude. Nevertheless, as a first cut, we code a prosecutor as vulnerable if he faced an opponent who garnered at least 10% of the vote in the previous election, and separately evaluate the announcement's effect on this subgroup of prosecutors and their 'safe' counterparts. As shown in Appendix Figure A7, both groups engaged in similar charging and sentencing behavior post-announcement.[33]

Alternative Explanations

In this section, we briefly consider some alternative explanations for our results.

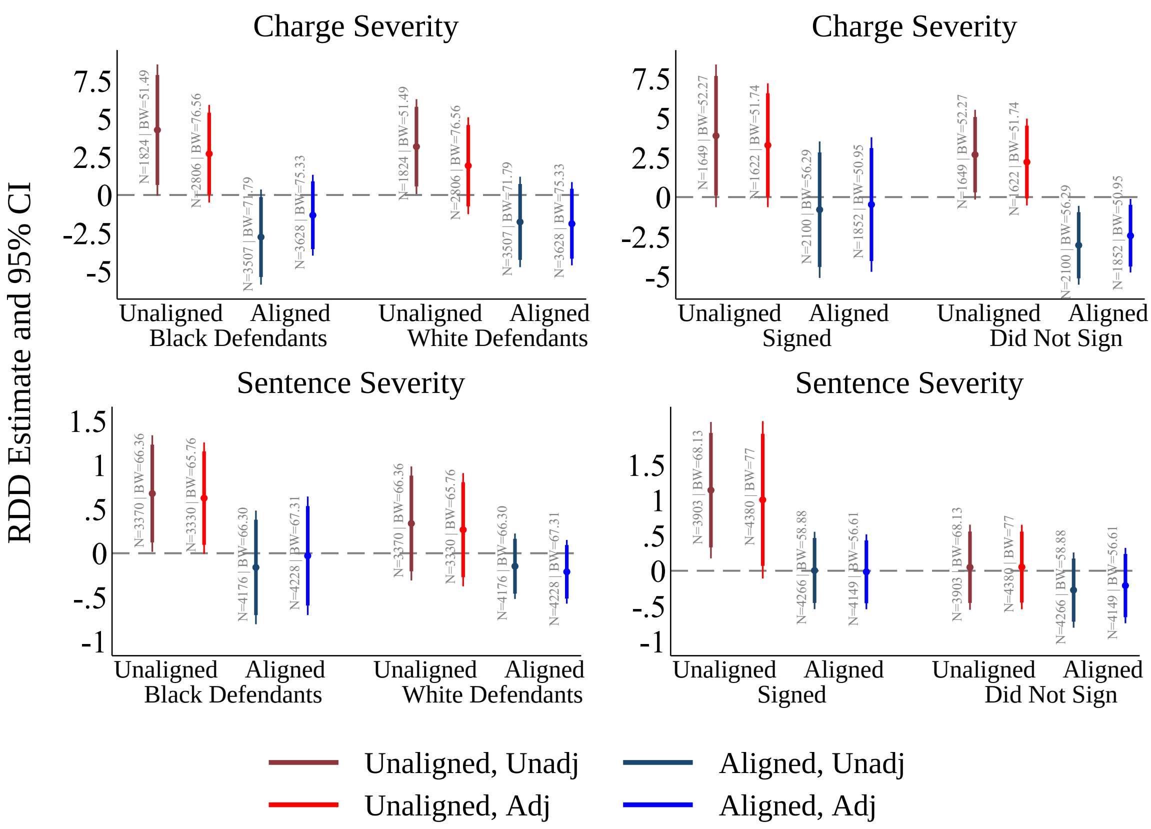
Public Opinion Prior evidence suggests that prosecutors may alter their charging behavior to align with local-level public opinion (Nelson, 2014). Is it possible that the shifts we observe in charging and sentencing behavior simply reflect post-announcement changes in public support for disenfranchisement? It seems unlikely: a wholesale change in public opinion regarding disenfranchisement should have motivated increases or decreases in prosecutor punitiveness *across the board*, rather than generating differential effects by alignment status.

Racial Attitudes and Disenfranchisement A more plausible possibility is that our results are generated by the *interaction* of the announcement and cross-district differences in racial attitudes. Our results could obtain, for example, if prosecutors aligned with the Democratic party reduced sentences assigned to (aligned) African-American defendants post-announcement, to demonstrate their opposition to this historical disen-

[33]This finding does not depend on the 90% cutoff we selected.

franchisement, while their Republican-aligned counterparts *increased* sentences against (unaligned) African-American defendants post-announcement. We test the validity of this theory by analyzing the effect of McAuliffe's announcement on prosecutor charging and sentencing, disaggregating by both political alignment and race. The left-hand column of Figure 9 displays the results. We find that Democratic party-aligned prosecutors decrease charging and sentencing severity against African-American defendants, while increasing severity for white defendants, while Republican party-aligned prosecutors do the opposite.

Figure 9: Effect of Announcement by Defendant Race and Prosecutor Attitudes



Left panel: RDD estimates and 90/95% confidence intervals, disaggregated effect on charging (top) and sentencing (bottom) of African American and white defendants respectively. Right panel: RDD estimates and 90/95% confidence intervals, disaggregated effect on prosecutors who signed / did not sign an amicus brief opposing McAuliffe's executive order. BW refers to the MSE-optimal bandwidth; N to the number of observations within that bandwidth.

Prosecutor attitudes Another possibility is that, rather than responding to constituent preferences, prosecutors were adjusting charges and sentences to communicate their own attitudes about felon disenfranchisement. These views were varied, even within-party: after McAuliffe's announcement, a bipartisan group of prosecutors signed an amicus brief filed with the state supreme court opposing the executive order.[34] Could prosecutors' punitiveness have shifted simply according to their ideological views on felon disenfranchisement? To answer this question, we separately estimate the announcement's effect on prosecutors who signed the amicus brief and prosecutors who did not. Overall the results, shown in the right-hand panel of Figure 9, show that prosecutors who signed the brief and prosecutors who did not responded to the announcement in the same way, becoming more punitive towards unaligned defendants, and less punitive towards aligned defendants.

Discussion

Scholars interested in the strategic behavior of elected prosecutors have often focused on their incentives to improve conviction rates, sentencing outcomes, and trial wins in order to signal quality or competence to a fixed pool of voters. Such behavior would arguably be consistent with a prosecutor's responsibility, as an elected official, to perform the will of her constituents — albeit in tension with her obligation as an officer of the court to perform justice. Our findings complicate the story further, suggesting that prosecutors may also use the tools of their office to shape the voter pool and improve the chance of a favored electoral outcome.

Our findings also contribute to recent scholarship on the ways in which political actors may attempt to 'tilt' electoral outcomes in their own favor. Although existing work

[34] "Brief of Amici Curiae Commonwealth's Attorneys In Support Of Petitioners," in *Howell v. McAuliffe*. Available at https://www.brennancenter.org/sites/default/files/analysis/Howell%20v%20McAuliffe%20-%20Prosecutors%20Amicus.pdf

focuses on the incentives of state-level politicians to *construct* laws and electoral rules that bend outcomes towards themselves (Helmke, Kroeger and Paine, 2022; Eubank and Fresh, 2022), our findings suggest that the local officials who *implement* these laws may also play an important role. An interesting avenue for future research would be to explore the extent to which local-level and state-level actors have complementary, as opposed to competing, goals. For example, if government is divided at the state and local levels, laws written at the state level with a view to disenfranchising one group might be used, at the local level, to disenfranchise another.

Finally, a major implication of our paper is that the composition of the felon population may depend on the partisan identity of local criminal justice officials. More specifically, the assignment of felon status may be endogenous to the degree of political threat or benefit a defendant poses to a prosecutor. Further research might explore the implications of this endogeneity for broader studies of felon political behavior.

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Appendix

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Contents

A Model

Consider a criminal defendant D and a prosecutor, P. The prosecutor can offer D a plea sentence x_P. The defendant can accept it, or refuse it and go to trial at cost c. c is known to D, but P knows only that c is distributed on $U[\underline{c}, \overline{c}]$. (The uniform distribution is unnecessary but ensures a tractable closed form solution.)

At trial, the defendant will be convicted (given the evidence against her) with probability π. If she is convicted, sentence x_T is imposed.

To provide a baseline for analysis, begin by supposing that there are no felon disenfranchisement laws, and the prosecutor derives utility only from imposing sentences and going to trial; in particular, he gets x from imposing sentence x on the defendant, and pays a cost k from going to trial.

The defendant's utility is the sentence imposed on her plus the cost of trial, if trial occurs. Thus, the defendant gets $-x_P$ from taking the plea deal, $-x_T - c$ if convicted at trial, and $-c$ if acquitted. Putting these together, the defendant accepts the plea deal offered by the prosecutor if

$$x_P \leq \pi x_T + c.$$

P gets x from sentence x and pays k for trial. Then P's utility from setting x_P is:

$$Pr(x_P - \pi x_T \leq c)x_P + Pr(x_P - \pi x_T > c)(\pi x_T - k).$$

Since c is distributed uniformly on $[\underline{c}, \overline{c}]$, we can rewrite this

$$\frac{\overline{c} - (x_P - \pi x_T)}{\overline{c} - \underline{c}} x_P + \frac{x_P - \pi x_T - \underline{c}}{\overline{c} - \underline{c}}(\pi x_T - k).$$

Differentiating with respect to x_P the first order condition is

$$\frac{1}{\overline{c} - \underline{c}}(\overline{c} - 2x_P + 2\pi x_T - k) = 0,$$

so the optimal x_P^* is

$$x_P^* = \frac{\overline{c} + 2\pi x_T - k}{2},$$

and the prosecutor sets some $x_P^* > 0$ whenever the cost to him of trial is not so enormous as to make the risk of trial untenable. (We assume for the remainder of the model that the parameters are such that the prosecutor never offers a plea of 0.)

Adding felon disenfranchisement laws

Now suppose that P also cares about his own, or his copartisans' reelection, *and* that there is a felony disenfranchisement law on the books which affects the ability of convicted felons to vote. Suppose the prosecutor believes that an enfranchised defendant would vote for his opponent (or his copartisans' opponents). Let $\lambda \geq 0$ represents the payoff to the prosecutor from preventing the defendant from voting.

The felon disenfranchisement law may be structured in one of two ways. First, it may be set up so that for practical purposes, an individual is unable to vote upon simple conviction of a felony. This might include both cases where felons are explicitly permanently disenfranchised by law, and cases where to regain the franchise would be so costly or time consuming that few individuals attempt it. Second, the felon disenfranchisement law may be structured so that a felon is disenfranchised only for the length of her prison or probation sentence.

Convictions alone are disenfranchising

Begin with the case where a simple conviction is permanently disenfranchising. If the prosecutor expects the defendant to vote for his or his copartisans' opponents, his utility is:

$$Pr(x_P - \pi x_T \leq c)(x_P + \lambda) + Pr(x_P - \pi x_T > c)(\pi(x_T + \lambda) - k).$$

Differentiating with respect to x_P the FOC is now:

$$\frac{1}{\overline{c}-\underline{c}}(\overline{c}-2x_P+2\pi x_T-(1-\pi)\lambda-k)=0,$$

So the optimal x_P is:

$$x'_P=\frac{\overline{c}+2\pi x_T-(1-\pi)\lambda-k}{2}.$$

The optimal plea sentence x_P under a felon disenfranchisement law that disenfranchises felons upon conviction is *lower* than the optimal plea sentence with no felon disenfranchisement law: $x'_P<x^*_P$. The reason is that compared to the baseline, the prosecutor now values conviction more relative to sentence, so he offers better plea deals in order to ensure (disenfranchising) conviction.

Felony sentences are disenfranchising for the length of the sentence

Consider an alternative regime in which convicted felons are disenfranchised only for the duration of their sentence. Now, we can think of sentence length as affecting the probability that the defendant can vote in the next election. In particular, suppose that the probability of being disenfranchised in the next election is increasing in sentence length. To make this concrete, suppose there is a maximum possible sentence of life, x_L, and all sentences x are in the interval $[0,x_L]$. Let the probability of being disenfranchised in the next election be $\frac{x}{x_L}$, so that it is 0 when the sentence is 0, increasing in the length of the sentence, and reaches 1 when the sentence assigned reaches x_L. Assume that $\lambda\leq x_L$, i.e., the prosecutor does not value disenfranchising the defendant for some length of time, more than he would value giving the defendant life in prison.

Now the prosecutor's utility is

$$Pr(x_P-\pi x_T\leq c)(x_P+\lambda\frac{x_P}{x_L})+Pr(x_P-\pi x_T>c)(\pi(x_T+\lambda\frac{x_T}{x_L})-k).$$

Differentiating yields the first order condition,

$$\frac{1}{\overline{c}-\underline{c}}\left(1+\frac{\lambda}{x_L}\right)(\overline{c}-2x_P+2\pi x_T)-k=0.$$

The prosecutor's optimal x_P is now

$$x''_P=\frac{\overline{c}+2\pi x_T}{2}-\frac{k}{2(1+\frac{\lambda}{x_L})}$$

Comparing the optimal plea offer in this disenfranchisement regime, x''_P, with the optimal offers in the two regimes above, it is clear that $x'_P < x^*_P < x''_P$. In other words, a prosecutor who perceives a defendant's vote as disfavorable makes lower plea offers when disenfranchisement occurs via conviction, and higher plea offers when disenfranchisement occurs only for the length of the sentence.

Felon disenfranchisement when the prosecutor values the defendant's vote

What if the prosecutor believes that the defendant he must attempt to convict and sentence would be likely to vote in a way *favorable* to him or his copartisans?

Begin with the situation where disenfranchisement is automatic upon conviction. In this scenario, conviction now comes at a cost, $-\lambda$. Assuming that the prosecutor cannot simply dismiss felony charges for individuals he believes would vote in a way he approves, the prosecutor's utility from setting x_P is now:

$$Pr(x_P-\pi x_T\leq c)(x_P-\lambda)+Pr(x_P-\pi x_T>c)(\pi(x_T-\lambda)-k)$$

Differentiating with respect to x_P the FOC is now:

$$\frac{1}{\overline{c}-\underline{c}}(\overline{c}-2x_P+2\pi x_T+(1-\pi)\lambda-k)=0$$

So

$$x'_P=\frac{\overline{c}+2\pi x_T+(1-\pi)\lambda-k}{2}$$

Comparing to the optimal plea offer in the baseline case where the prosecutor does not care about the disenfranchising consequences of conviction, observe that now $x'_P > x^*_P$: the prosecutor offers a higher plea. This is because the prosecutor wants to *avoid* conviction if possible. He therefore offers harsher plea deals in an attempt to force the defendant to trial, where with at least some probability she will be acquitted.

Suppose now that disenfranchisement depends only on the length of the sentence imposed, so that the probability the defendant is prevented from voting is (as before) equal to $\frac{x}{x_L}$ and increasing in sentence length.

Now the prosecutor's utility is

$$Pr(x_P - \pi x_T \leq c)(x_P - \lambda \frac{x_P}{x_L}) + Pr(x_P - \pi x_T > c)(\pi(x_T - \lambda \frac{x_T}{x_L}) - k)$$

Differentiating yields the first order condition:

$$\frac{1}{\overline{c} - \underline{c}} \left(1 - \frac{\lambda}{x_L}\right) (\overline{c} - 2x_P + 2\pi x_T) - k = 0$$

The prosecutor's optimal x_P is now

$$x''_P = \frac{\overline{c} + 2\pi x_T - \frac{k}{1 - \frac{\lambda}{x_L}}}{2}$$

Now the prosecutor wants to offer very lenient plea deals in order to maximize the probability that the defendant can vote for him, so we have:

$$x''_P < x^*_P < x'_P$$

In a case where the prosecutor perceives the defendant's vote as favorable, he makes the lowest plea offers under a sentence-dependent disenfranchisement regime, and the highest plea offers under a conviction-dependent disenfranchisement regime.

A hybrid regime?

Notice that one might have a hybrid regime. Suppose that the prosecutor does not want the defendant to be able to vote, and that for some types of defendant, conviction alone is permanently disenfranchising, but for others, disenfranchisement only lasts as long as the sentence imposed on the defendant. More specifically, suppose with probability $1-p$ the defendant will be permanently disenfranchised upon conviction, and with p she will be disenfranchised only for the length of her sentence. Then the prosecutor sets x_P to maximize the p-weighted average of the two utility functions above, and the optimal x_P^* is now a function of p:

$$x_P^*(p) = \frac{\overline{c} + 2\pi x_T}{2} - \frac{(1-p)(1-\pi)\lambda + k}{2(1 + p\frac{\lambda}{x_L})}.$$

Observe that as $p \to 0$, $x_P^*(p) \to x'_P$ and as $p \to 1$, $x_P^*(p) \to x''_P$.

Now any change in the law that leads to an increase in p, i.e. that increases the probability that a given felon will be disenfranchised only for the length of her sentence, will lead to an increase in the optimal sentence x_P^* (and the converse is true if the prosecutor wants the defendant to vote). To see this observe that the second term in the left hand side expression above is decreasing in p. Since the term is negatively signed, the whole expression is increasing in p. Equivalently:

$$\frac{\partial x_P^*}{\partial p} = \frac{2\left(1 + p\frac{\lambda}{x_L}\right)[(1-\pi)\lambda + k] + 2\frac{\lambda}{x_L}\left((1-p)(1-\pi)\lambda + k\right)}{4(1 + p\frac{\lambda}{x_L})^2} > 0.$$

Defendant values the right to vote

So far, we have assumed that the defendant does not take into account her potential disenfranchisement when choosing between accepting a plea deal and going to trial. We believe this assumption is plausible for a variety of reasons: individuals may be ignorant of the consequences of a felony conviction for their voting rights, they may feel that disenfranchisement is a very small cost, relative to a prison sentence, or they may heavily discount the cost of future disenfranchisement relative to costs and benefits in the present.

However, in this section we incorporate a disutility to the defendant from suffering disenfranchisement, and briefly show that this does not affect our results unless the disutility to the defendant from disenfranchisement is sufficiently large, relative to the benefit to the prosecutor.

Suppose the defendant is disenfranchised upon conviction and suffers $-\alpha > 0$ from disenfranchisement. Then her payoff from taking a plea becomes $-x_P - \alpha$, her payoff from conviction at trial becomes $-x_T - \alpha - c$, and her payoff from acquittal remains $-c$. She now accepts plea x_P if

$$-x_P - \alpha \leq \pi(-x_T - \alpha) - c,$$

or rearranging, if

$$c \geq x_P - \pi x_T + (1-\pi)\alpha$$

Prosecutor likes disenfranchising defendant Assuming the prosecutor likes disenfranchising the defendant, the prosecutor's utility is now:

$$Pr(x_P - \pi x_T + (1-\pi)\alpha \leq c)(x_P + \lambda) + Pr(x_P - \pi x_T + (1-\pi)\alpha > c)(\pi(x_T + \lambda) - k)$$

And the optimal x_P is now:

$$x_P'^{\alpha} = \frac{\overline{c} + 2\pi x_T - (1-\pi)(\lambda + \alpha) - k}{2}.$$

Notice that here, the defendant requires an additional discount in the plea offer to be willing to accept it, because she dislikes disenfranchisement. Consequently, the optimal plea offer is even lower here than in the case where the defendant is not concerned with disenfranchisement, $x_P'^{\alpha} < x_P' < x_P^*$.

Now suppose disenfranchisement is only for the length of the sentence. Recalling the prosecutor's payoff in this case, suppose the defendant suffers a parallel disutility, such that after a plea she receives $-x_P - \alpha\frac{x_P}{x_L}$, or $-x_P(1+\frac{\alpha}{x_L})$, and after conviction at trial she receives $-x_T(1+\frac{\alpha}{x_L})$. We assume $\alpha \leq x_L$, i.e., the defendant cannot suffer more from disenfranchisement for some length of time than she would suffer from life in prison.

She pleads guilty now if

$$c \geq \left(1 + \frac{\alpha}{x_L}\right)(x_P - \pi x_T)$$

and plugging in and solving for the optimal x_P, the prosecutor now chooses:

$$x_P''^{\alpha} = \frac{\overline{c} + 2\left(1+\frac{\alpha}{x_L}\right)2\pi x_T}{2\left(1+\frac{\alpha}{x_L}\right)} - \frac{k}{2\left(1+\frac{\lambda}{x_L}\right)}.$$

It is clear that this plea offer is smaller than the one the prosecutor would make (x_P'' above) if he wanted to disenfranchise the defendant and the defendant was unconcerned with disenfranchisement. Thus, the defendant's concern for disenfranchisement has a downward effect on the plea offer made by the prosecutor. However, a comparison of this plea offer ($x_P''^{\alpha}$) to the plea offer the prosecutor would make if he were not desirous of disenfranchising the defendant (x_P^*), reveals that the former is still larger than the latter, so long as α is not too large. In particular, the prosecutor's concern for disenfranchising the defendant via a longer sentence continues to motivate him to increase his plea offer so long as

$$\alpha < \frac{\lambda k}{\overline{c} + \frac{\lambda}{x_L}(\overline{c} - k)}.$$

For α below this threshold, we have:

$$x_P^* < x_P''^{\alpha} < x_P''.$$

Note that the α threshold below which felon disenfranchisement motivates the prosecutor to impose harsher plea deals on defendants he wants to disenfranchise, is increasing in the benefit the prosecutor obtains from disenfranchisement λ, and the cost to the prosecutor of trial k.

Prosecutor wants defendant to vote Now consider the situation where there is a disenfranchisement law and the prosecutor *wants* the defendant to vote.

Suppose conviction is disenfranchising. Recall that when the defendant does not care about disenfranchisement, the prosecutor sets

$$x'_P = \frac{\overline{c} + 2\pi x_T + (1-\pi)\lambda - k}{2},$$

which is *larger* than the optimal plea offer in the baseline model, because the prosecutor wants to deter the defendant from taking the plea.

If the defendant cares α about disenfranchisement, she is already to some extent deterred from taking the plea. The consequence is that here, the prosecutor does not need to set such a harsh plea offer to induce optimal deterrence. Specifically, he now sets:

$$x_P^{'\alpha} = \frac{\overline{c} + 2\pi x_T + (1-\pi)(\lambda-\alpha) - k}{2}.$$

Notice that so long as $\alpha < \lambda$, this plea offer is still larger than the optimal plea $x_P^* = \frac{\overline{c}+2\pi x_T - k}{2}$ that the prosecutor would set absent a felon disenfranchisement law.

Finally, suppose the prosecutor wants the defendant to vote, and disenfranchisement only lasts for the duration of the sentence.

Then the prosecutor chooses

$$x_P^{''\alpha} = \frac{\overline{c} + 2\left(1+\frac{\alpha}{x_L}\right)2\pi x_T}{2\left(1+\frac{\alpha}{x_L}\right)} - \frac{k}{2\left(1-\frac{\lambda}{x_L}\right)}.$$

Recall that when the prosecutor wants the defendant to vote, and the defendant is unconcerned with disenfranchisement, the prosecutor chooses $x''_P < x^*_P$, where $x''_P = \frac{\overline{c}+2\pi x_T - \frac{k}{1-\frac{\lambda}{x_L}}}{2}$. A comparison reveals that $x_P^{''\alpha} < x''_P$. Thus, here, the defendant's concern for disenfranchisement pushes the prosecutor's optimal plea below what it would have been had only the prosecutor worried about disenfranchising the defendant.

B Additional Results

B.1 Main Results, Robustness

B.1.1 Excluding Independent Prosecutors

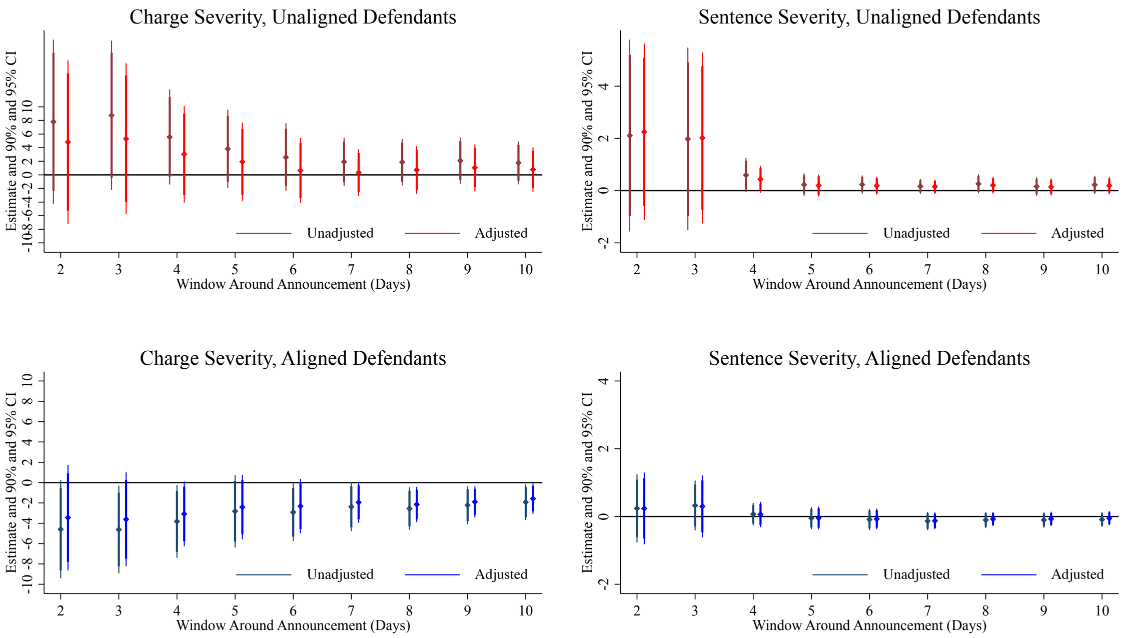
In this section we replicate the results presented in Table 2, Figure 5, and Figure 6, limiting our analysis to partisan prosecutors. Excluding Independents reduces the number of districts by half, from 118 to 61. The results are consistently signed, although some are weaker.

Table A1: Effect of McAuliffe's Order, By Alignment Status, Partisan Prosecutors

	Unaligned Defendants		Aligned Defendants		Unaligned Defendants		Aligned Defendants	
	Panel A. Charge Severity				*Panel B. Sentence Severity*			
RD Estimate	2.735	1.303	-2.676	-2.289	0.267	0.139	-0.174	-0.118
	(1.716)	(1.698)	(1.192)	(1.113)	(0.436)	(0.419)	(0.251)	(0.252)
Left Intercept	16.032	15.919	18.679	18.734	0.912	0.912	0.805	0.798
	(0.833)	(0.866)	(0.985)	(0.909)	(0.117)	(0.113)	(0.196)	(0.182)
Bandwidth	62.470	53.178	72.730	67.243	44.774	43.069	62.590	60.027
Eff. Observations	3775	3277	6640	6245	2451	2417	5535	5413
District FE	Y	Y	Y	Y	Y	Y	Y	Y
Covariates	N	Y	N	Y	N	Y	N	Y

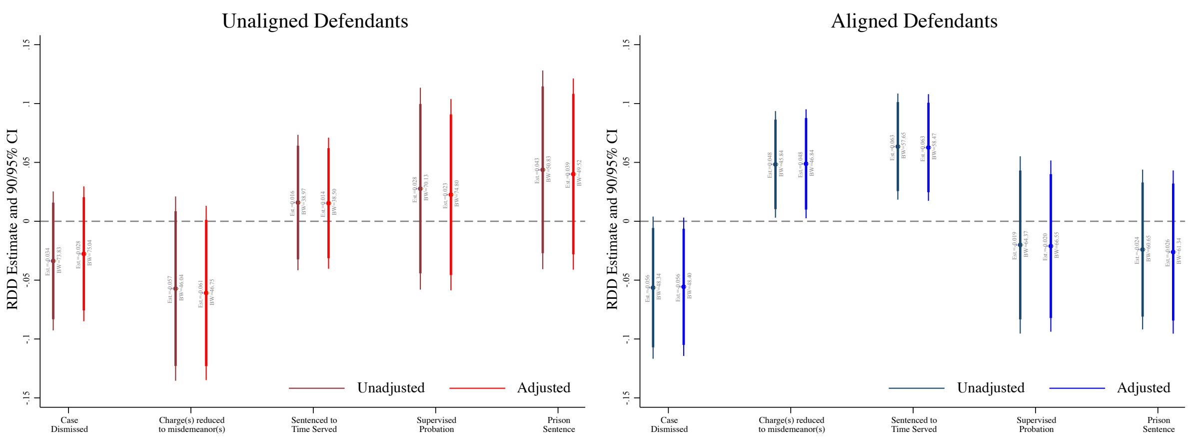
Charge and sentence severity shown in years. Left Intercept (left-hand side intercept) is the estimated outcome immediately before the announcement. For each defendant type, first columns include district-fixed effects; second columns also condition on gender and crime type. Robust standard errors clustered at the district-level. Only African-American and white defendants included (98% of all defendants).

Figure A1: Effect of Announcement, Pooled Difference-in-Means, Partisan Prosecutors



Point estimates and 90/95% confidence intervals for 2-10 day windows around the announcement. All estimates adjusted for district fixed effects; 'adjusted' results adjusted for pretreatment covariates. Standard errors clustered at the district. Clusters for charge severity: 25-51 for unaligned; 35-52 for aligned. Sentence severity clusters: 35-55 for aligned; 25-46 for unaligned.

Figure A2: Effect of Announcement on Sentencing Outcomes, Partisan Prosecutors



RD estimates and 90/95% robust confidence intervals for the post-announcement change in the probability that defendants' charges are (from left to right) dismissed, reduced from felonies to misdemeanors, sentenced to time served, sentenced to supervised probation, or sentenced to prison. All estimates adjust for district fixed effects; "adjusted" estimates also control for pre-treatment covariates.

B.1.2 All Voters Aligned with Democrats

Is it possible that in strongly Democratic districts, such as those bordering the Washington D.C. metro area, Democratic prosecutors perceive both African-American and white defendants as aligned? As we argue in the main text of our paper, even if in these districts the average white *citizen* is more likely than not to lean Democrat, the substantial demographic differences between the average white citizen and the average white felony defendant make it likely that the latter is still more likely than not to lean Republican. However, in this subsection, we examine this possibility further by replicating our main results in Table 2, recoding white citizens in heavily Democratic districts (districts where more than 70% of the vote went for Obama in 2012) as also-aligned. The results, shown below, are consistently signed, but generally weaker.

Table A2: Effect of McAuliffe's Order, By Alignment Status, All defendants coded 'aligned' in districts where > 70% voted for Barack Obama in 2012

	Unaligned Defendants		Aligned Defendants		Unaligned Defendants		Aligned Defendants	
	Panel A. Charge Severity				*Panel B. Sentence Severity*			
RD Estimate	1.987	1.391	-0.981	-0.615	0.526	0.532	-0.163	-0.151
	(1.229)	(1.216)	(1.033)	(1.035)	(0.330)	(0.311)	(0.178)	(0.181)
Left Intercept	17.098	17.070	18.429	18.437	0.778	0.775	0.844	0.829
	(0.720)	(0.741)	(0.790)	(0.754)	(0.090)	(0.093)	(0.142)	(0.136)
Bandwidth	67.534	52.201	78.270	61.642	54.828	54.930	72.418	70.448
Eff. Observations	5610	4408	9392	7521	4365	4365	8468	8300
District FE	Y	Y	Y	Y	Y	Y	Y	Y
Covariates	N	Y	N	Y	N	Y	N	Y

Charge and sentence severity shown in years. Left Intercept is the estimated outcome immediately before the announcement. For each defendant type, first columns include district-fixed effects; second columns also condition on gender and crime type. Robust standard errors clustered at the district-level. Only African-American and white defendants included (98% of all defendants).

B.1.3 Probation Violations

In this subsection we replicate the sentencing estimates in Table 2, including probation violation sentences. The results are similar.

Table A3: Announcement's Effect on Sentence, Including Probation Violation Charges

	Unaligned Defendants		Aligned Defendants	
RD Estimate	0.467	0.410	-0.149	-0.118
	(0.257)	(0.258)	(0.188)	(0.182)
Left Side Intercept	0.750	0.751	0.852	0.859
	(0.084)	(0.085)	(0.149)	(0.134)
Bandwidth	55.810	54.916	72.251	74.102
Eff. Observations	4907	4812	8066	8312
District FE	Y	Y	Y	Y
Covariates	N	Y	N	Y

Sentence severity shown in years. Left Intercept is the estimated outcome immediately before the announcement. For each defendant type, first columns include district fixed effects; second columns also condition on gender and crime type. Robust standard errors clustered at the district-level. Only African-American and white defendants included (98% of all defendants).

B.1.4 Local Randomization Inference

In this subsection we replicate Table 2 using local randomization inference.

Table A4: Effect of McAuliffe's Order on Charge and Sentence Severity By Alignment Status, Local Randomization Difference-in-Means Estimates

	Aligned Defendants				Unligned Defendants			
W	Diff. Means	Finite $P > \|T\|$	Asympt $P > \|T\|$	Eff. Obs.	Diff. Means	Finite $P > \|T\|$	Asympt $P > \|T\|$	Eff. Obs.
	Panel A. Charge Severity (Districts Present In Both Windows)							
1	-2.140	0.362	0.312	122	5.361	0.032	0.029	130
2	-3.231	0.110	0.100	168	4.195	0.066	0.061	173
3	-1.815	0.292	0.272	241	4.201	0.022	0.015	284
4	-3.064	0.014	0.015	432	3.665	0.026	0.013	367
5	-2.998	0.002	0.007	509	3.622	0.008	0.008	430
	Panel B. Sentence Severity: Districts Present In Both Windows							
1	0.435	0.240	0.220	113	1.579	0.218	0.125	81
2	0.309	0.316	0.280	113	1.385	0.264	0.188	99
3	-0.013	0.986	0.953	368	0.791	0.074	0.099	258
4	-0.115	0.662	0.558	603	0.390	0.178	0.233	430
5	-0.097	0.616	0.613	682	0.344	0.176	0.162	529

Table presents the window used (column 1); the observed difference in means for this window (columns 2 and 5); the p-value for the difference generated via a complete randomization procedure using 1000 randomly drawn randomizations—i.e., the proportion of random assignments to treatment and control for which the test statistic generated was more extreme than that actually observed—(columns 3 and 6); the p-value for the difference corresponding to a test of the average null hypothesis (columns 4 and 7), and the effective number of observations used (column 8). Districts included only if present both before/after announcement and range from 9 (some one-day windows) to 53 (some five-day windows).

B.2 Effects on Sentencing

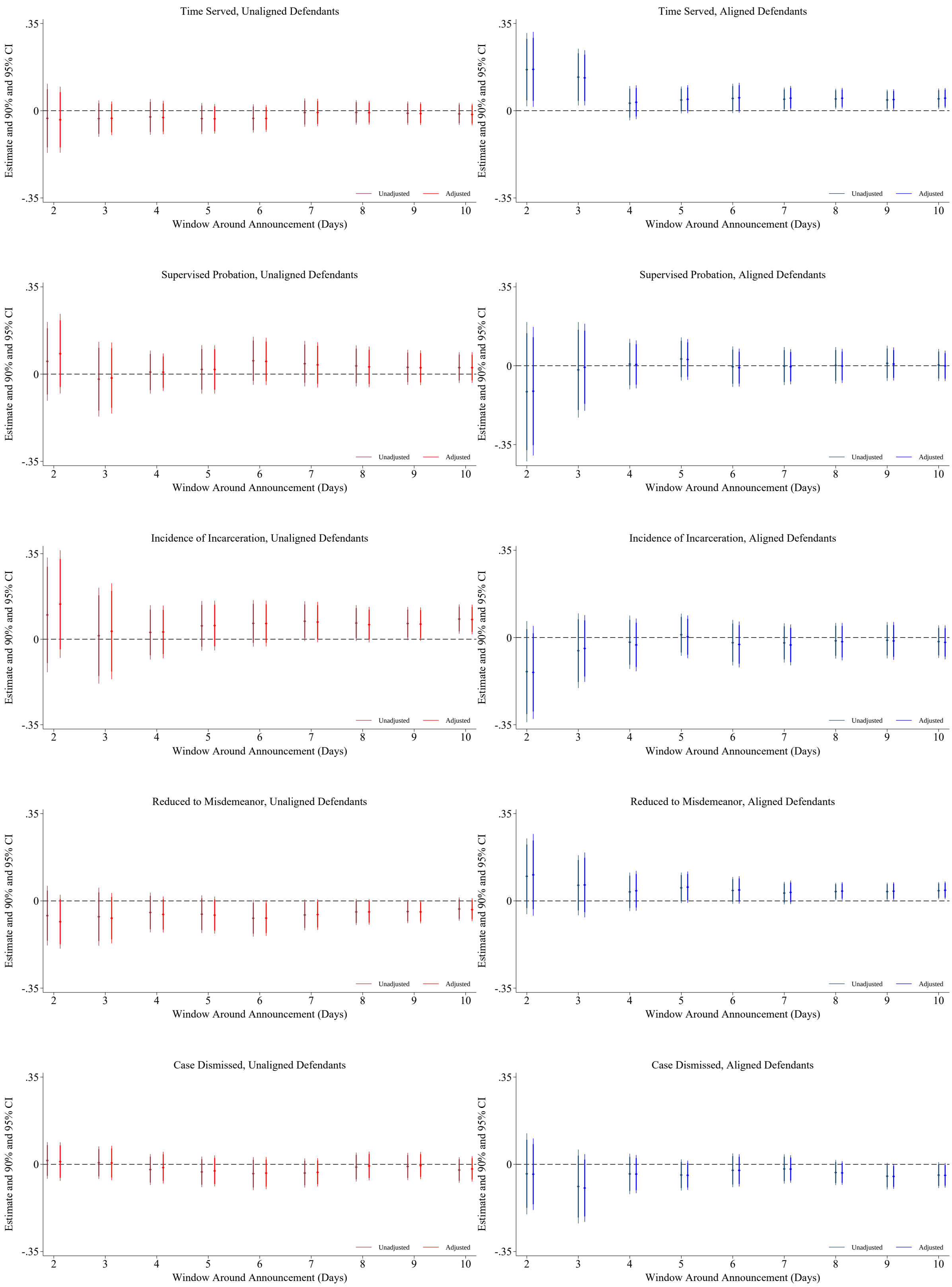
In this subsection we present estimates of the announcement's affect on sentencing, limiting our analysis to cases filed prior to the announcement. We also present a series of difference-in-means estimates of the announcement's effect on prosecutorial sentencing strategies.

Table A5: Effect of Announcement on Sentence Length (in years) Cases Filed Before Announcement

	Aligned Defendants				Unaligned Defendants			
RD Estimate	-0.173	-0.135	-0.104	-0.084	0.475	0.412	0.326	0.303
	(0.192)	(0.186)	(0.205)	(0.187)	(0.255)	(0.257)	(0.283)	(0.267)
Intercept	0.867	0.865	0.801	0.847	0.758	0.758	0.759	0.759
	(0.152)	(0.136)	(0.152)	(0.131)	(0.086)	(0.086)	(0.079)	(0.079)
Bandwidth	71.909	73.896	62.040	71.407	56.160	55.096	53.730	53.858
Eff. Obs.	7610	7787	6708	7601	4770	4685	4531	4531
District FE	Y	Y	Y	Y	Y	Y	Y	Y
Covariates	N	Y	N	Y	N	Y	N	Y
Ind. Sev.	N	N	Y	Y	N	N	Y	Y

Sentence severity shown in years. Left Intercept is the estimated outcome immediately before the announcement. For each defendant type, first columns include district-fixed effects; second columns also condition on gender and crime type. Robust standard errors clustered at the district level. Only African-American and white defendants included (98% of all defendants).

Figure A3: Effect of Announcement on Sentencing Outcomes, Difference-in-Means



Pooled estimates, 90/95% cluster-robust CIs for change in probability of time served, probation, prison, reduction to misdemeanor, or dismissal for unaligned/aligned defendants. All estimates adjust for district FEs; adjusted estimates adjust for gender/crime type. Clusters from 54-97 for aligned, 62-98 for unaligned defendants.

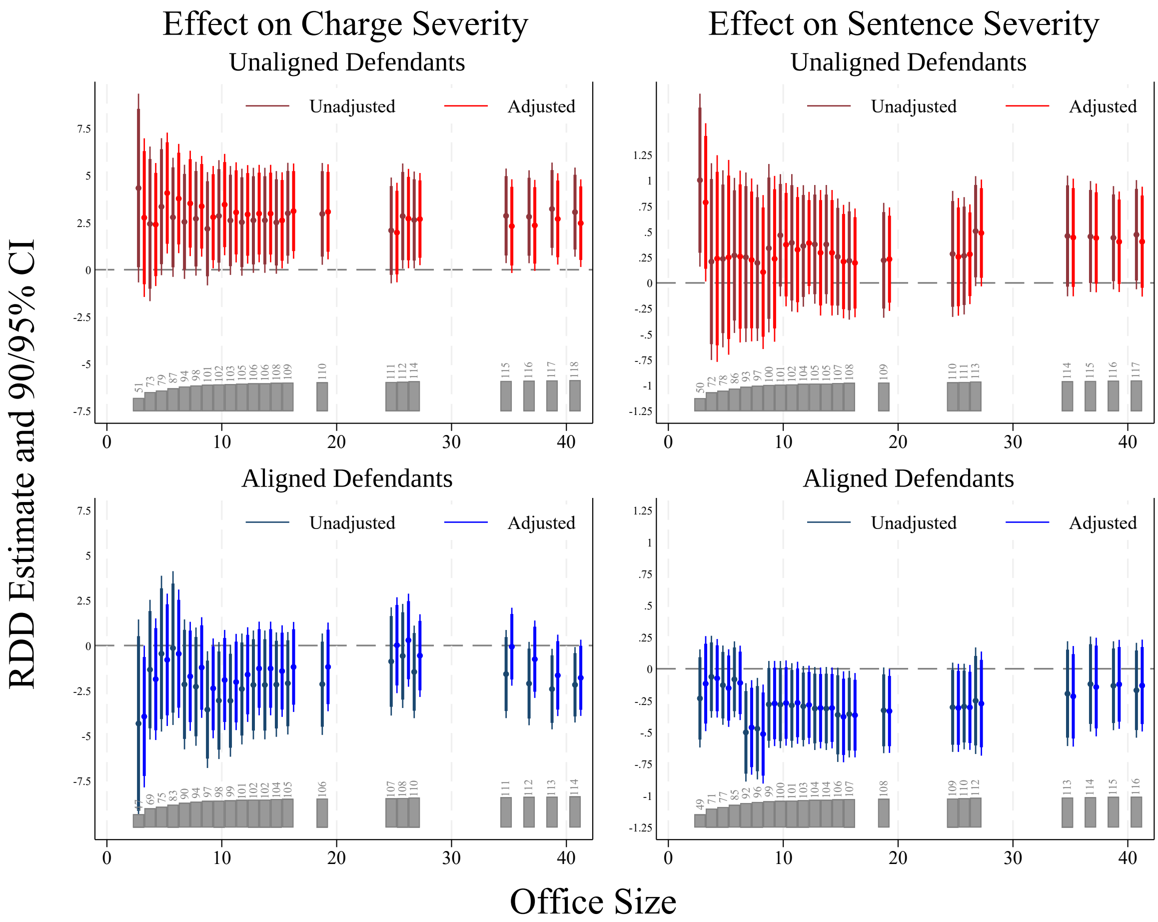
Table A6: Effect of McAuliffe's Order on Incidence of Time Served Sentence, By Alignment Status, Local Randomization Difference-in-Means Estimates

	Aligned Defendants				Unaligned Defendants			
W	Diff. Means	Finite $P > \|T\|$	Asympt $P > \|T\|$	Eff. Obs.	Diff. Means	Finite $P > \|T\|$	Asympt $P > \|T\|$	Eff. Obs.
2	0.093	0.144	0.080	153	0.007	1.000	0.912	99
3	0.025	0.536	0.458	378	-0.009	1.000	0.815	258
4	0.047	0.110	0.086	603	-0.014	0.752	0.646	430
5	0.046	0.076	0.066	682	-0.015	0.638	0.598	529

Column 1 lists the window used for inference, Columns 2 and 5 the observed difference; in means for each window; Columns 3 and 6 the p-value corresponding to a sharp null hypothesis test and generated via a complete randomization procedure using 1000 randomly drawn randomizations. Columns 4 and 7 present the p-value corresponding to a test of the average null hypothesis. Columns 5 and 9 list the effective number of observations used. Only districts with at least one observation in both windows included.

B.3 Heterogeneous Effects

Figure A4: Effect on Charge and Sentence Severity, by Office Size

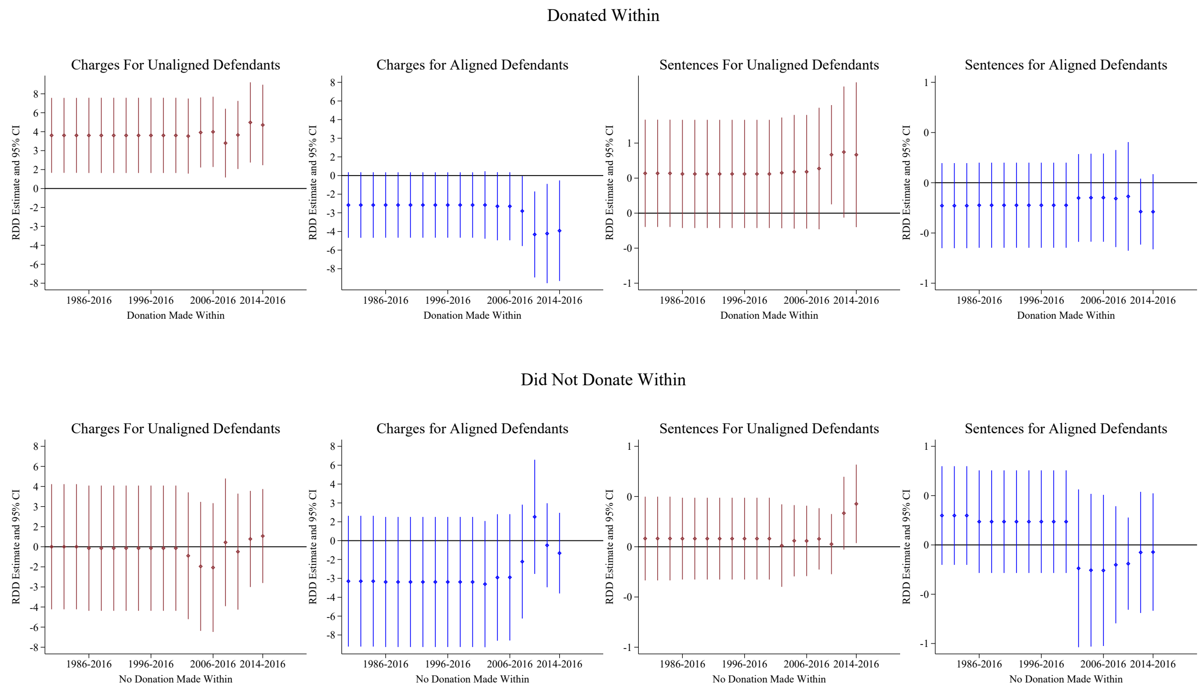


The plots in Figure A4 present the effect of the announcement on charge severity (top row) and sentence severity (bottom row) for aligned (right column) and unaligned (left column) defendants, disaggregating by office size (X-axis). Specifically, the coefficient estimates within each plot represent the effect of the announcement for all offices at or under a given size, beginning at an office size of 2 and increasing to the maximum office size of 41. In general, the results do not vary greatly by office size, especially for the unaligned. For aligned defendants,the announcement's effect on sentence severity seems to be driven primarily by medium-sized offices. We obtain estimates of prosecutor office size from a survey conducted by the Virginia Compensation Board in 2018 and presented in their memorandum, "Workgroup Study of the Impact of Body Worn Cameras on Workload in Commonwealth's Attorneys' Offices," available at https://www.scb.virginia.gov/docs/bodycameraworkgroupreport.pdf.

B.4 Personal or Partisan Incentives

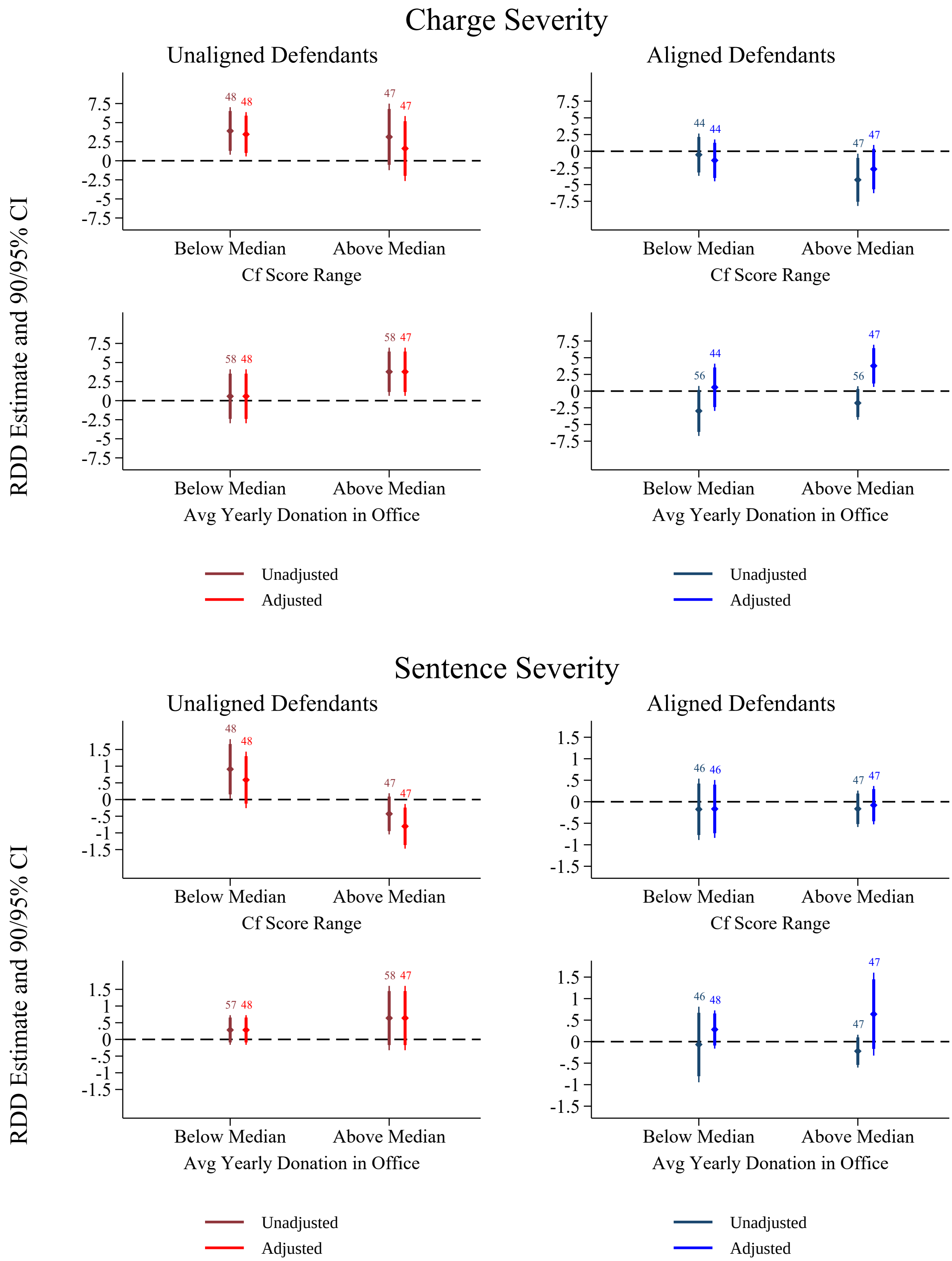
In this section we present results on prosecutors' personal political incentives, as well as their more general partisan motivations.

Figure A5: Announcement Effect By Recency of Last Donation (No Covariates)



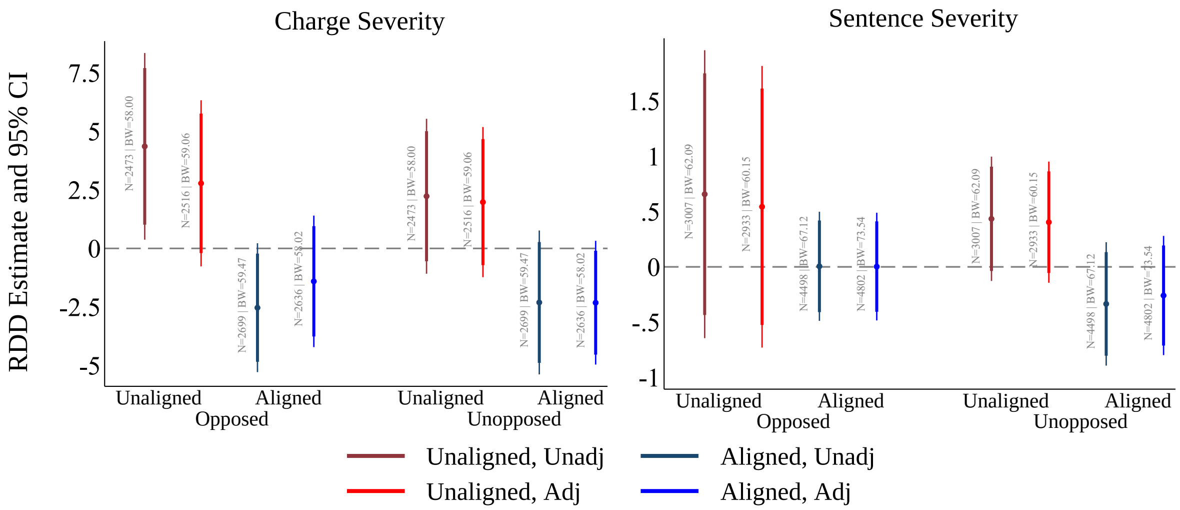
Point estimates and 95% confidence intervals, adjusted for district fixed effects. Clusters vary across specifications. For the announcement's effect on charges, cluster size ranges from 78 to 41 among those who donated (depending on the interval) and from 65 to 33 among those who did not. For the announcement's effect on sentences, cluster size ranges from 81 to 46 for those who donated (depending on the interval), and from 35 to 67 for those who did not.

Figure A6: Announcement Effect, by CF Scores and Amounts



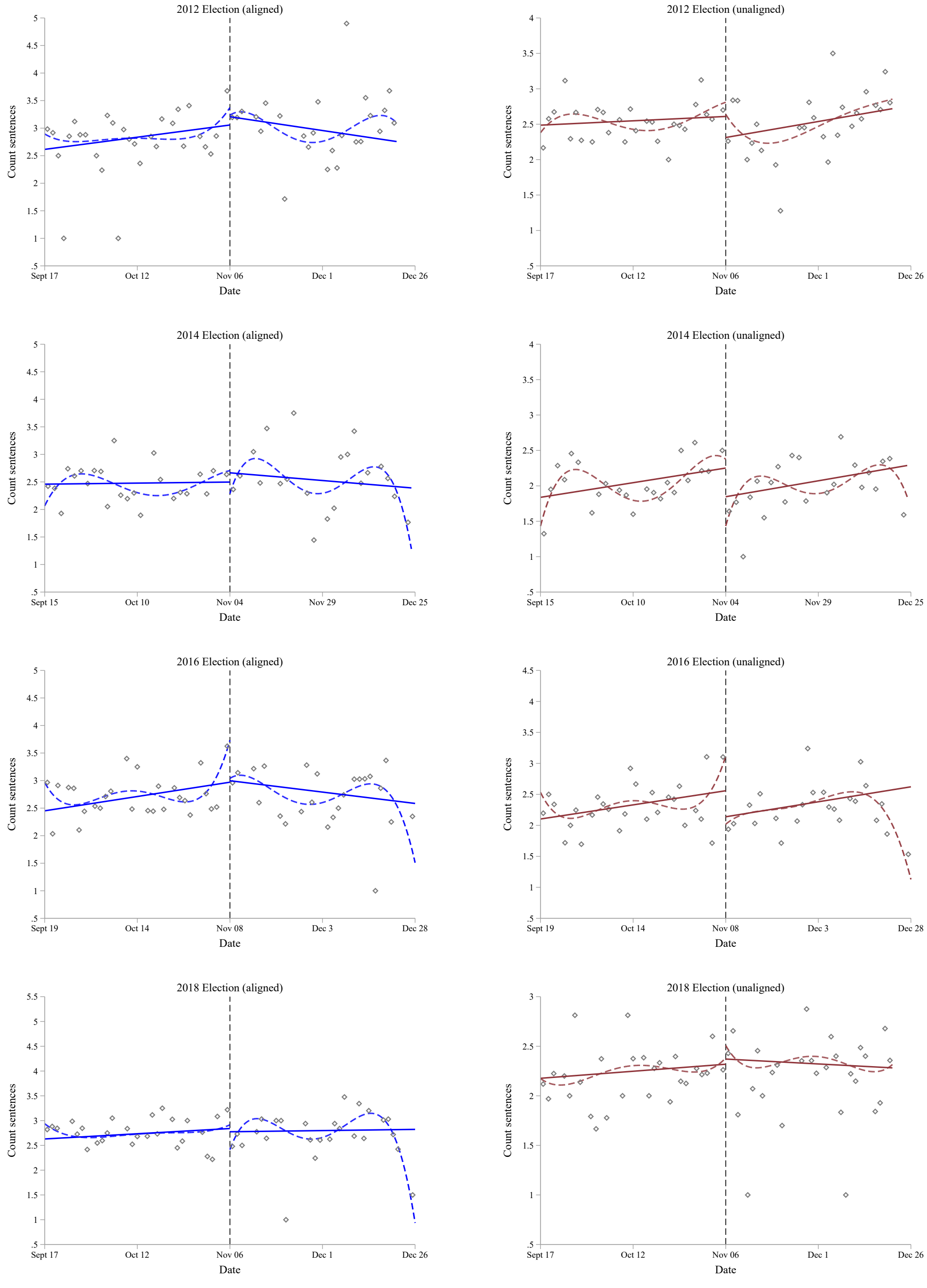
RDD estimates and associated 90 and 95% confidence intervals. All models adjust for district fixed effects. “Adjusted” specifications control for additional pretreatment covariates (crime type and gender). Cluster sizes vary across specification, and are displayed above the 95% confidence intervals.

Figure A7: Effects by Electoral Threat



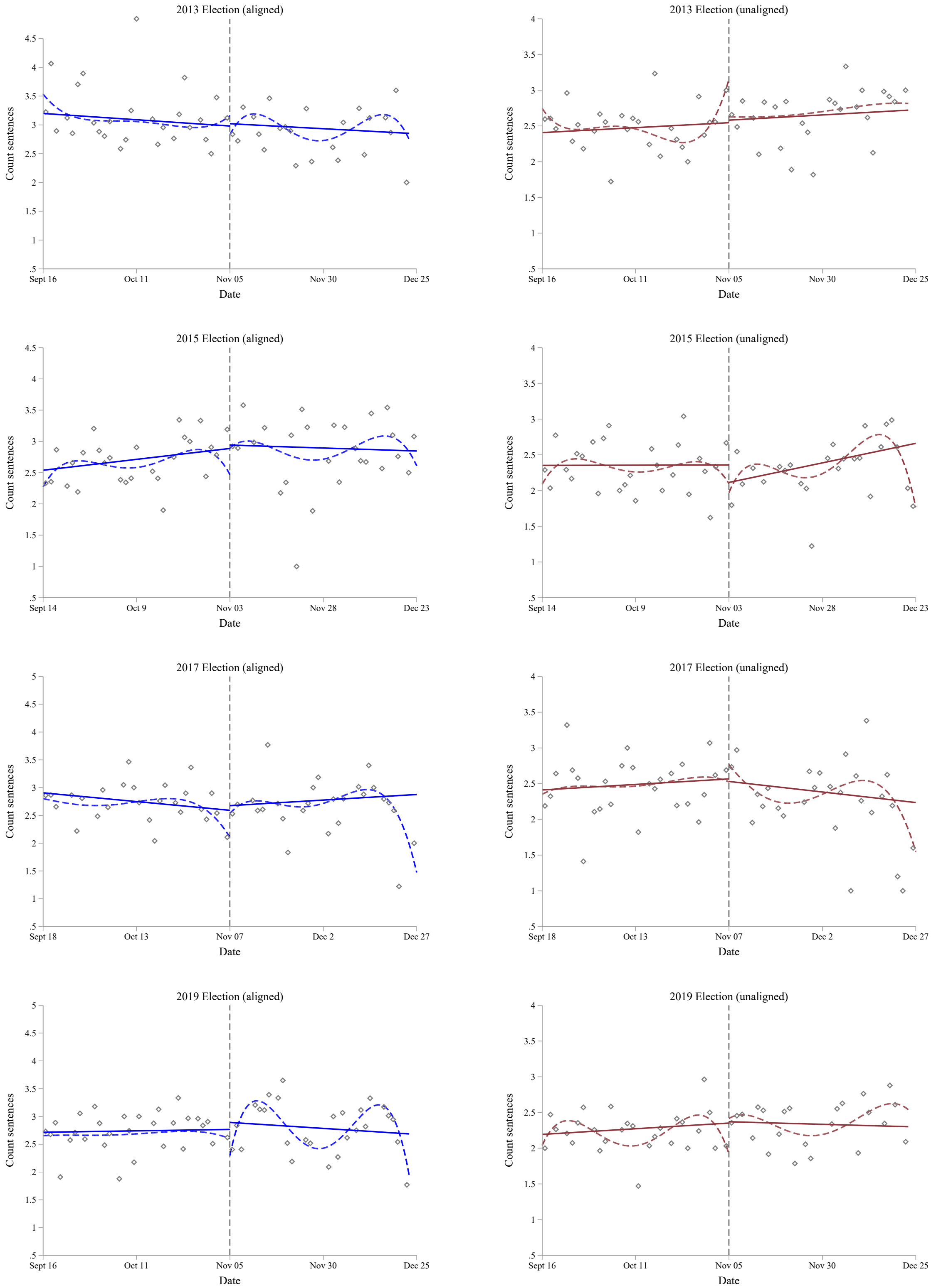
The figure presents separate estimates of the announcement's effect on charge severity (left panel) and sentence severity (right panel) in districts where the prosecutor was threatened in the last election (faced an opponent who won more than 10% of the vote) and those where the prosecutor was secure. BW refers to the MSE-optimal bandwidth, while N refers to the number of observations within that bandwidth.

Figure A8: Total cases disposed of for aligned (left) and unaligned (right) defendants before/after Federal election dates



Dashed lines are a kernel weighted polynomial of order 4 and solid lines are linear predictions.

Figure A9: Total cases disposed of for aligned (left) and unaligned (right) defendants before/after state election dates



Dashed lines are a kernel weighted polynomial of order 4 and solid lines are linear predictions.